SFU NatLangLab

CMPT 713: Natural Language Processing

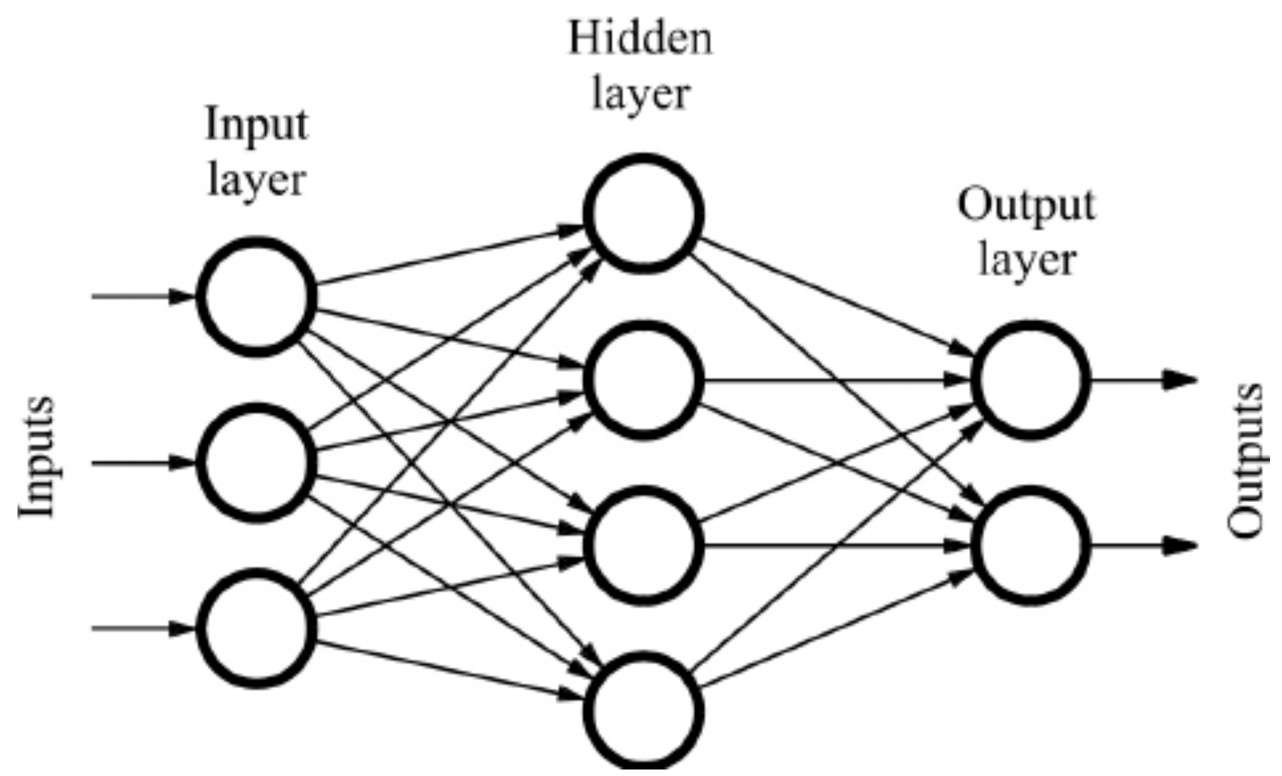
Neural Network Basics

Spring 2023
2023-01-19

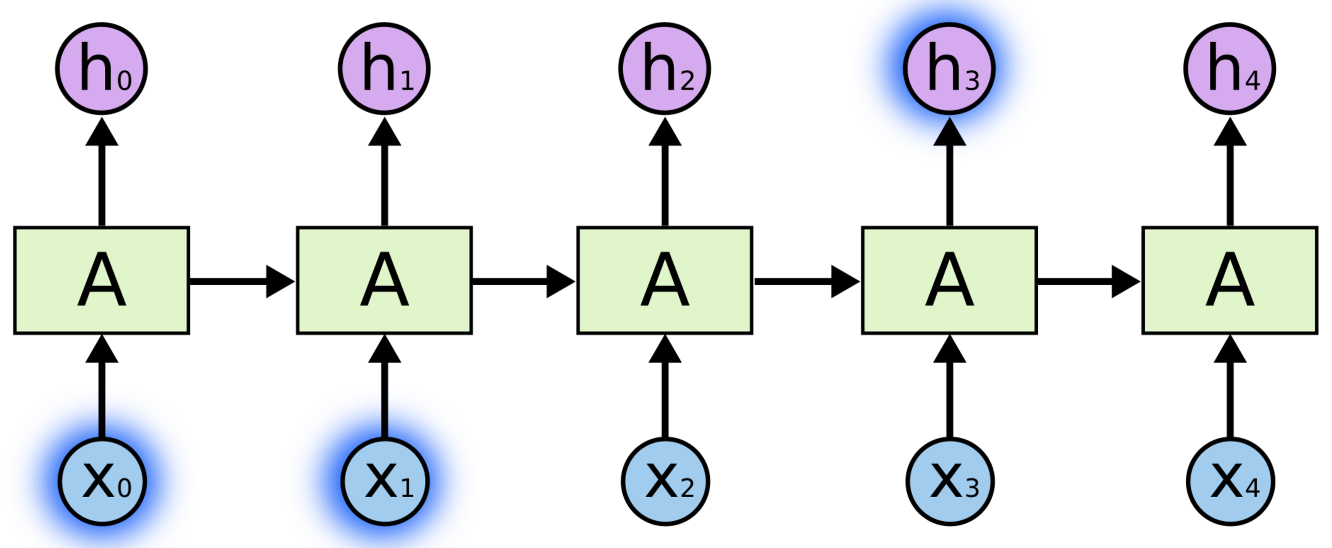
Adapted from slides from Danqi Chen and Karthik Narasimhan

Neural networks for NLP

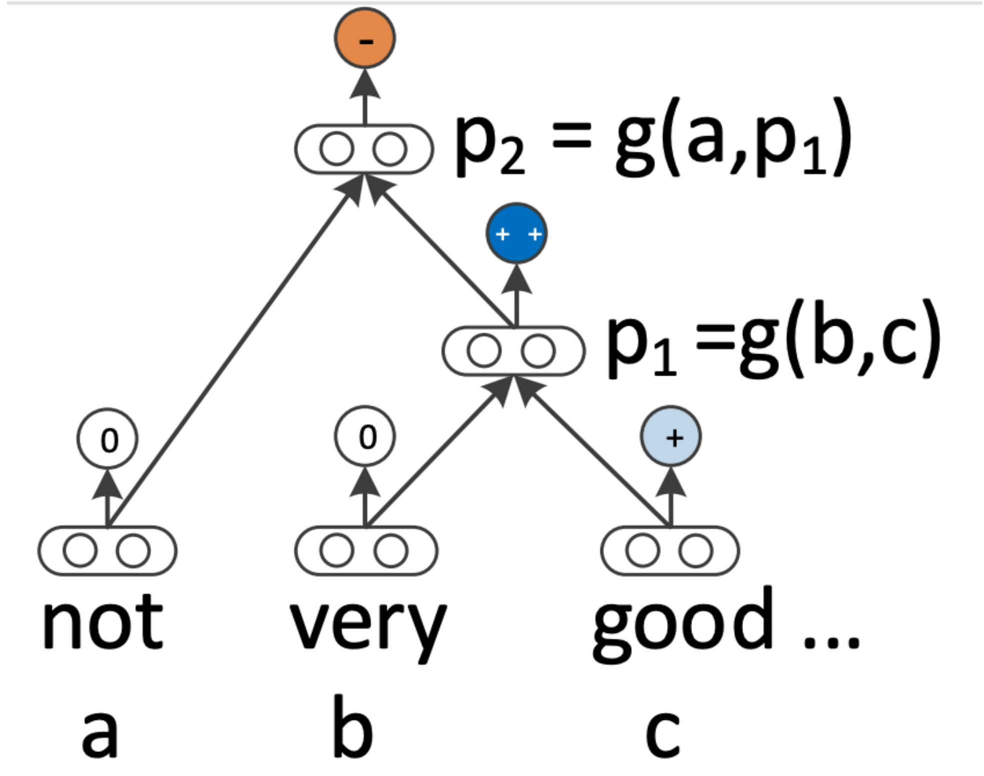
Feed-forward NNs



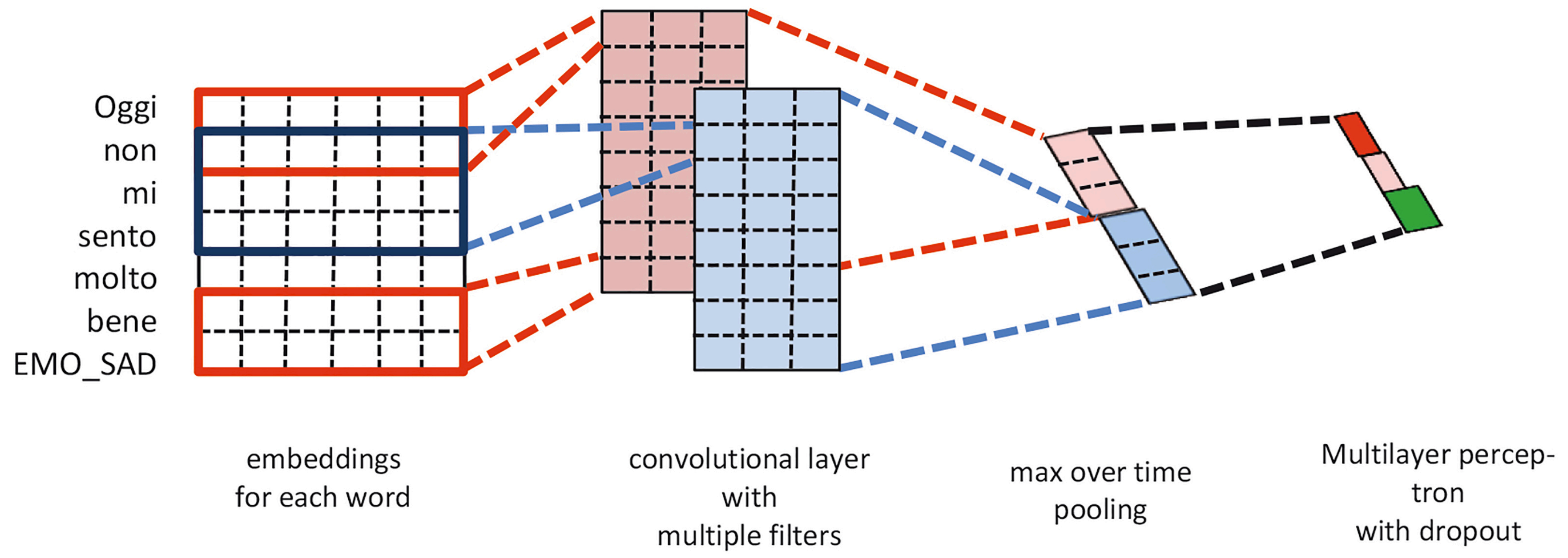
Recurrent NNs



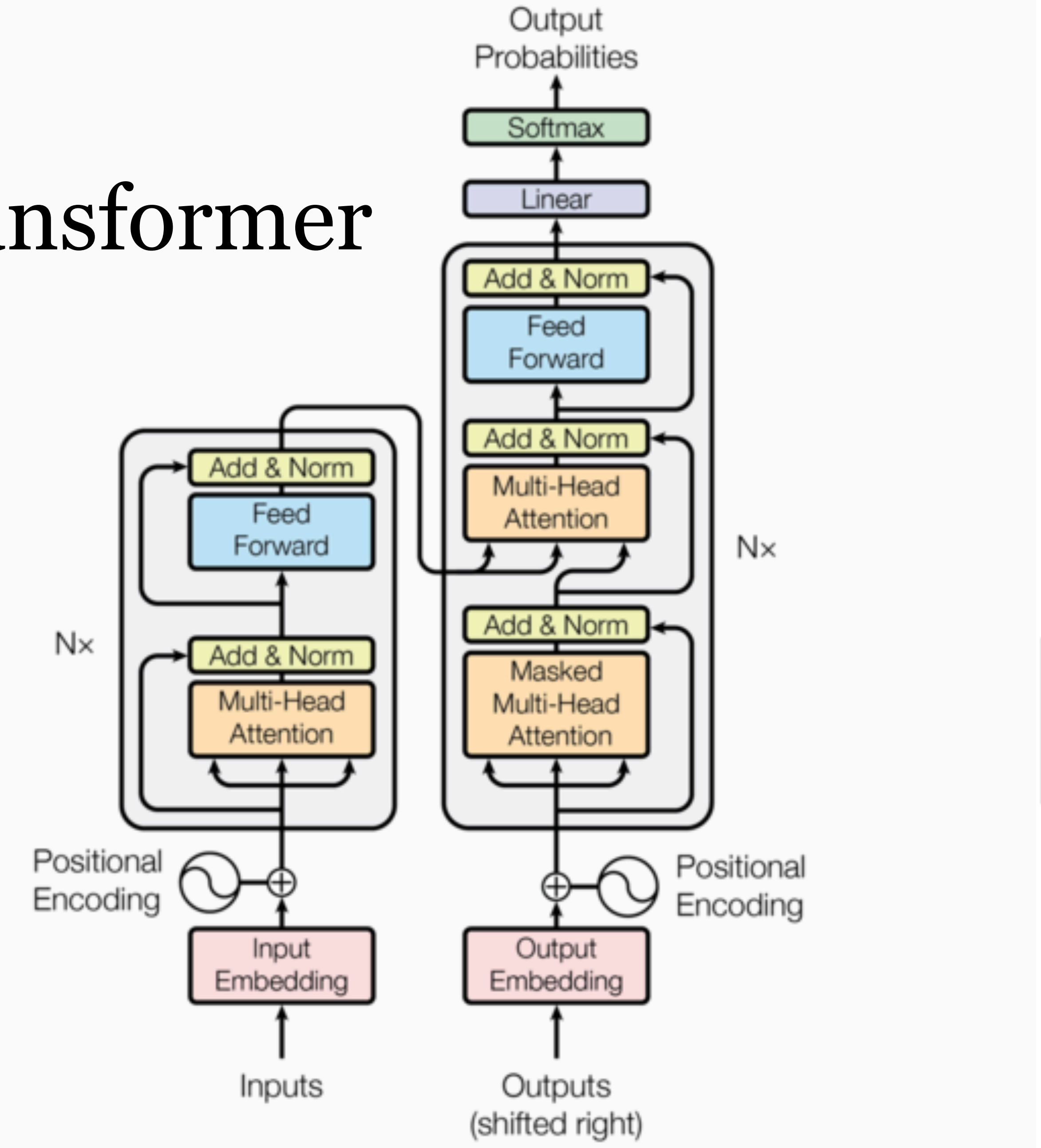
Recursive NNs



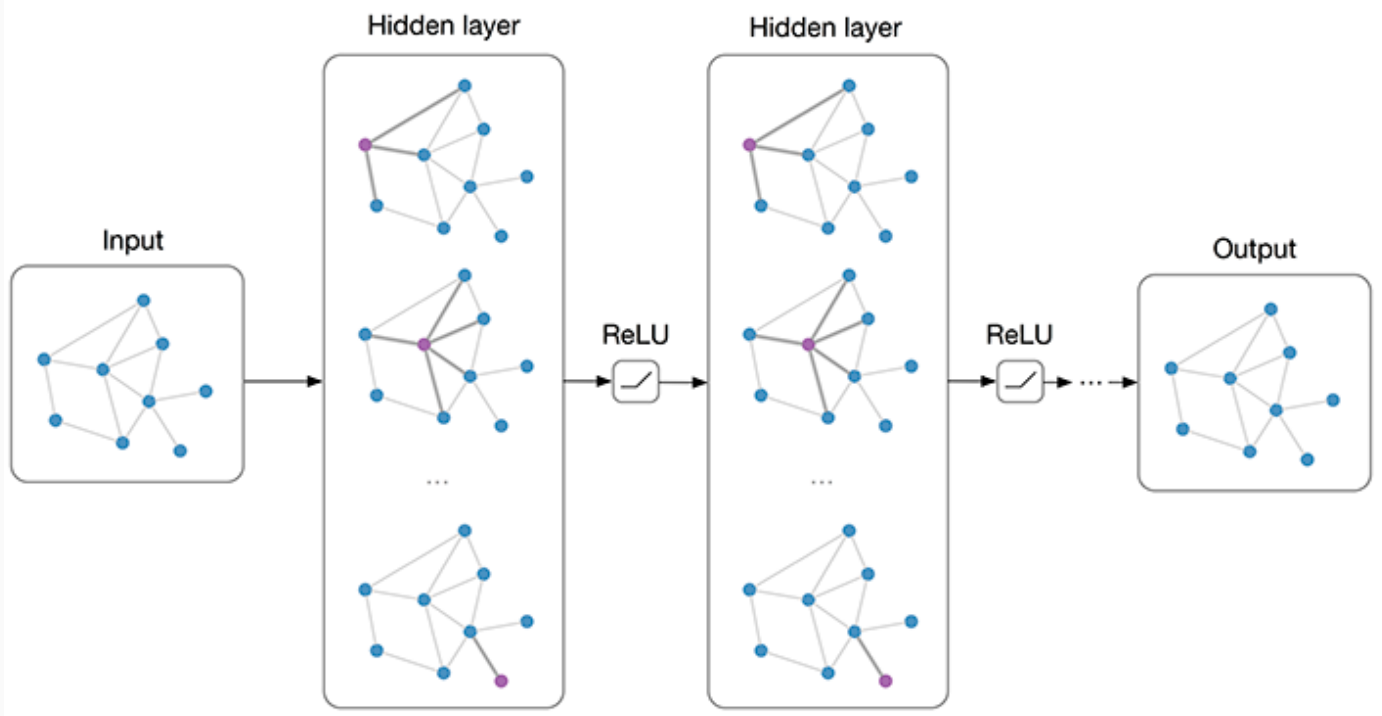
Convolutional NNs



Transformer



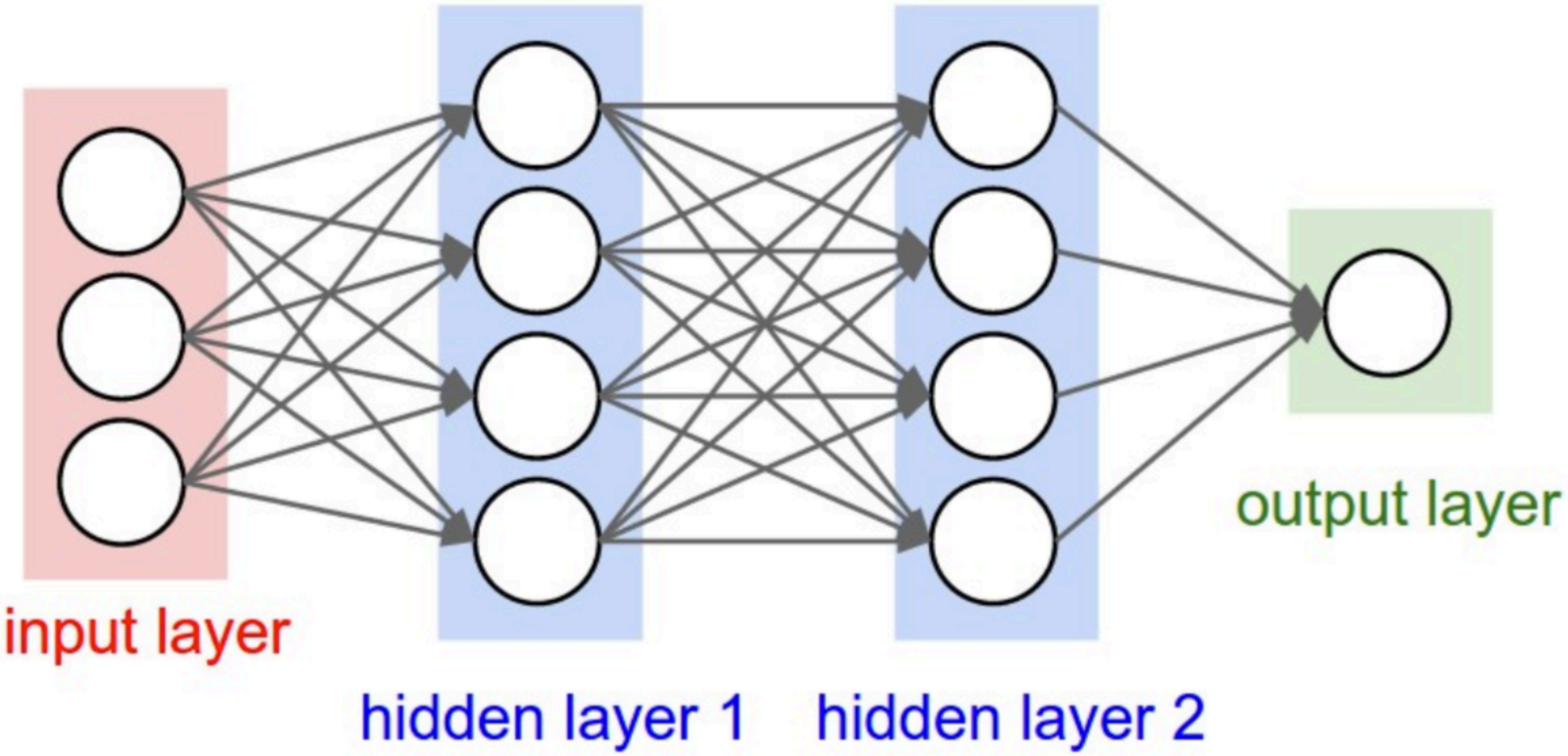
Graph NNs



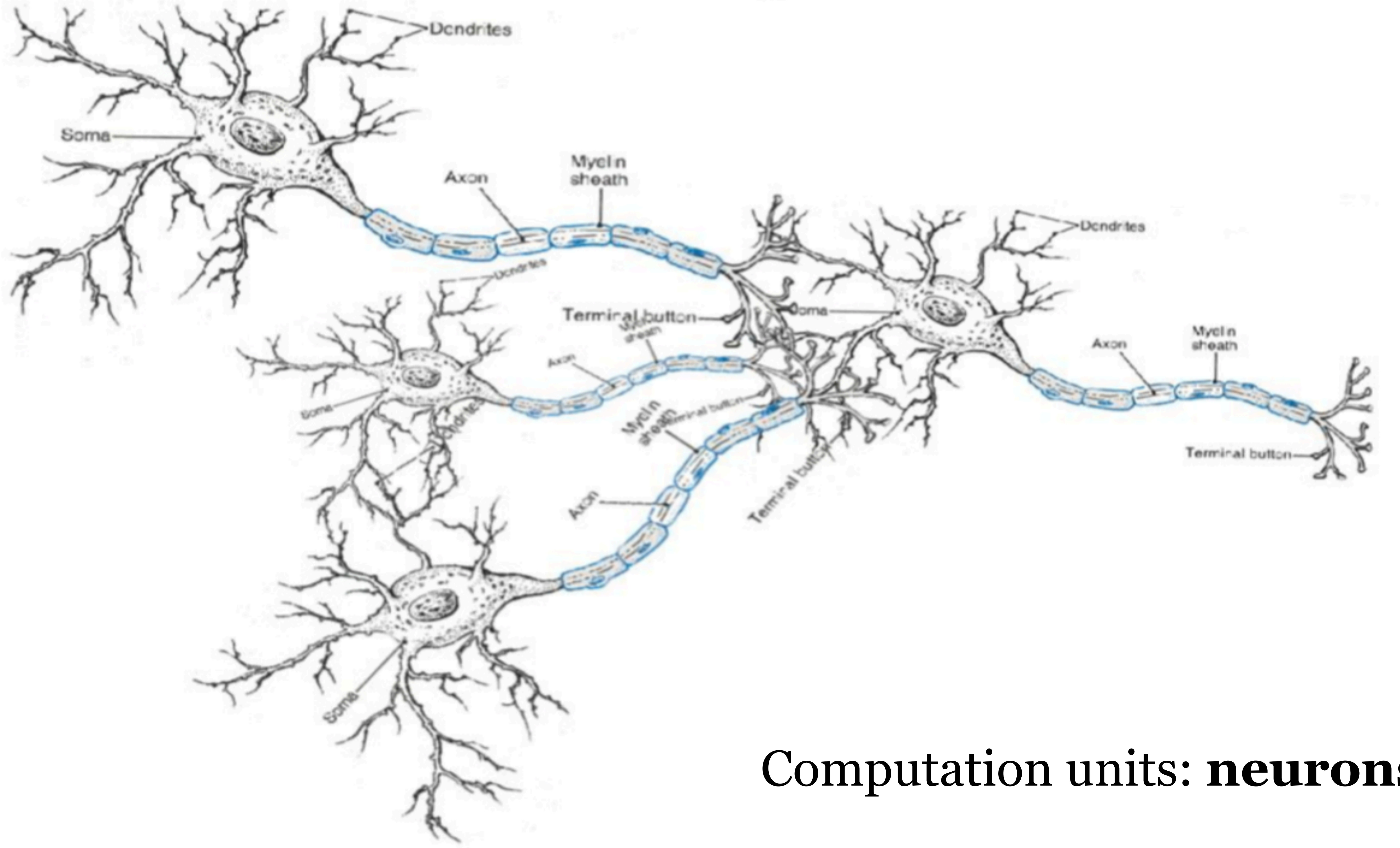
Always coupled with word embeddings…

Feed-forward NNs

- Input: $x_1, \ldots, x_d$
- Output: $y \in \{0,1\}$



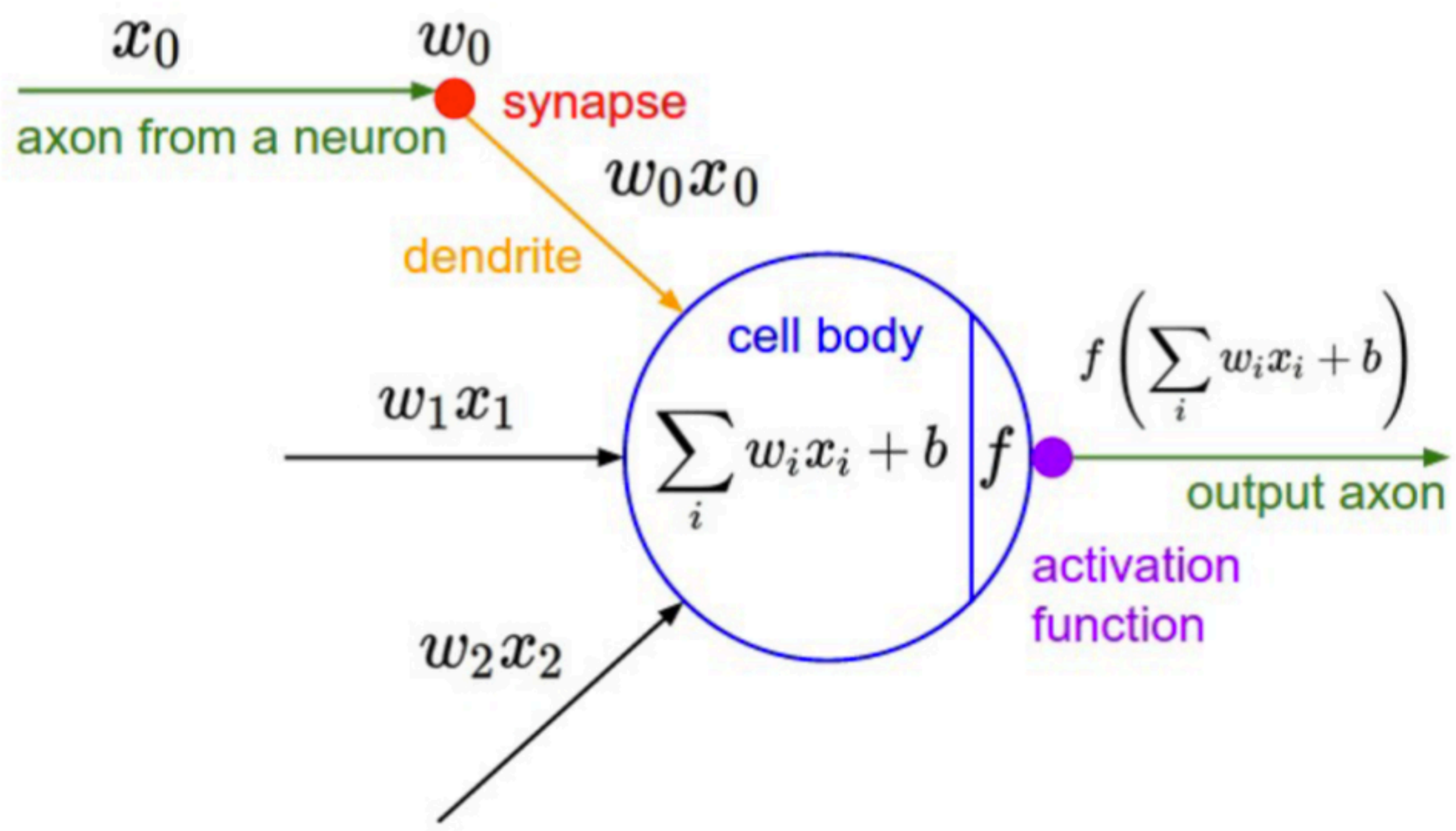
Neural computation



Computation units: **neurons**

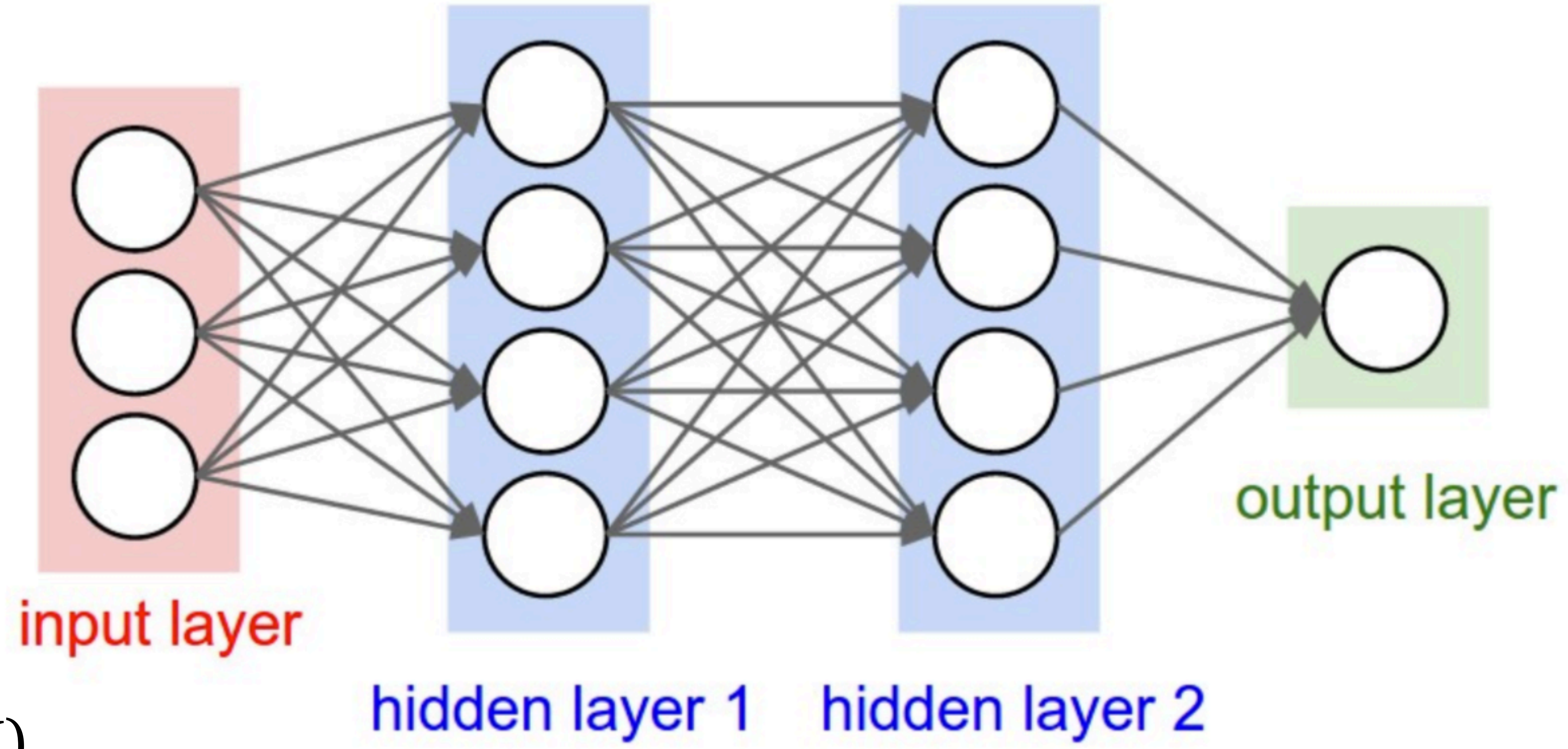
An artificial neuron

- A neuron is a computational unit that has scalar inputs and an output
- Each input has an associated weight.
- The neuron multiples each input by its weight, sums them, applied a **nonlinear function** to the result, and passes it to its output.



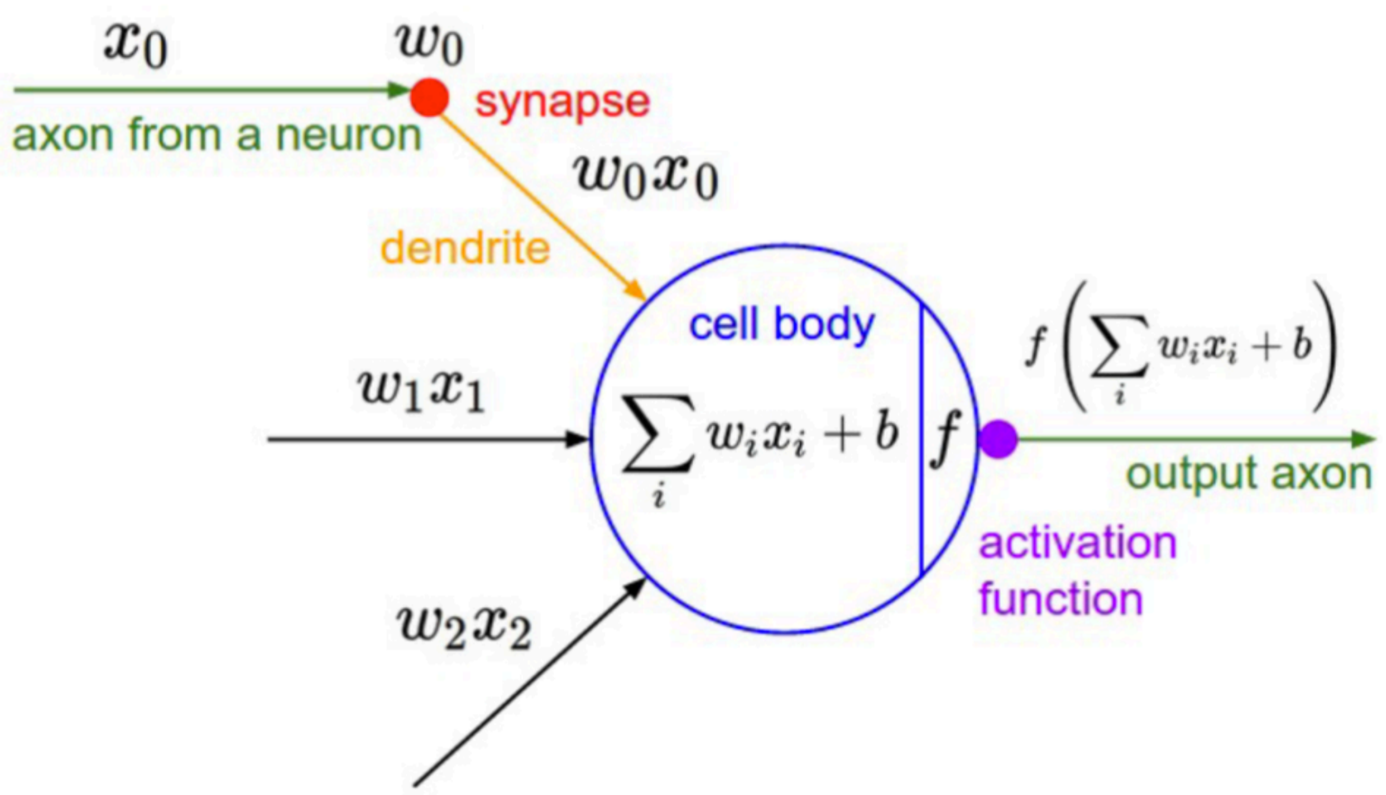
Neural networks

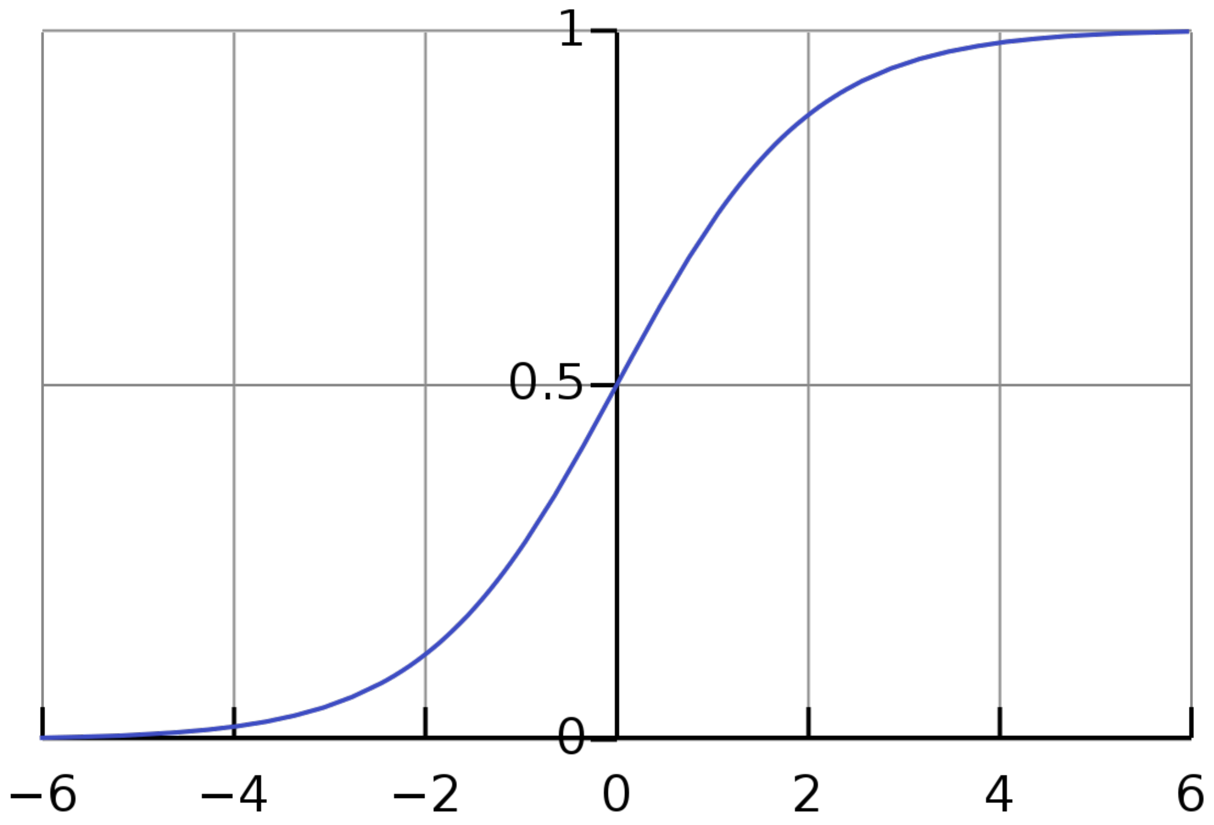
- The neurons are connected to each other, forming a **network**
- The output of a neuron may feed into the inputs of other neurons



- Feed forward network (FFN)
- Fully connected network (FCN)

A neuron can be a binary logistic regression unit

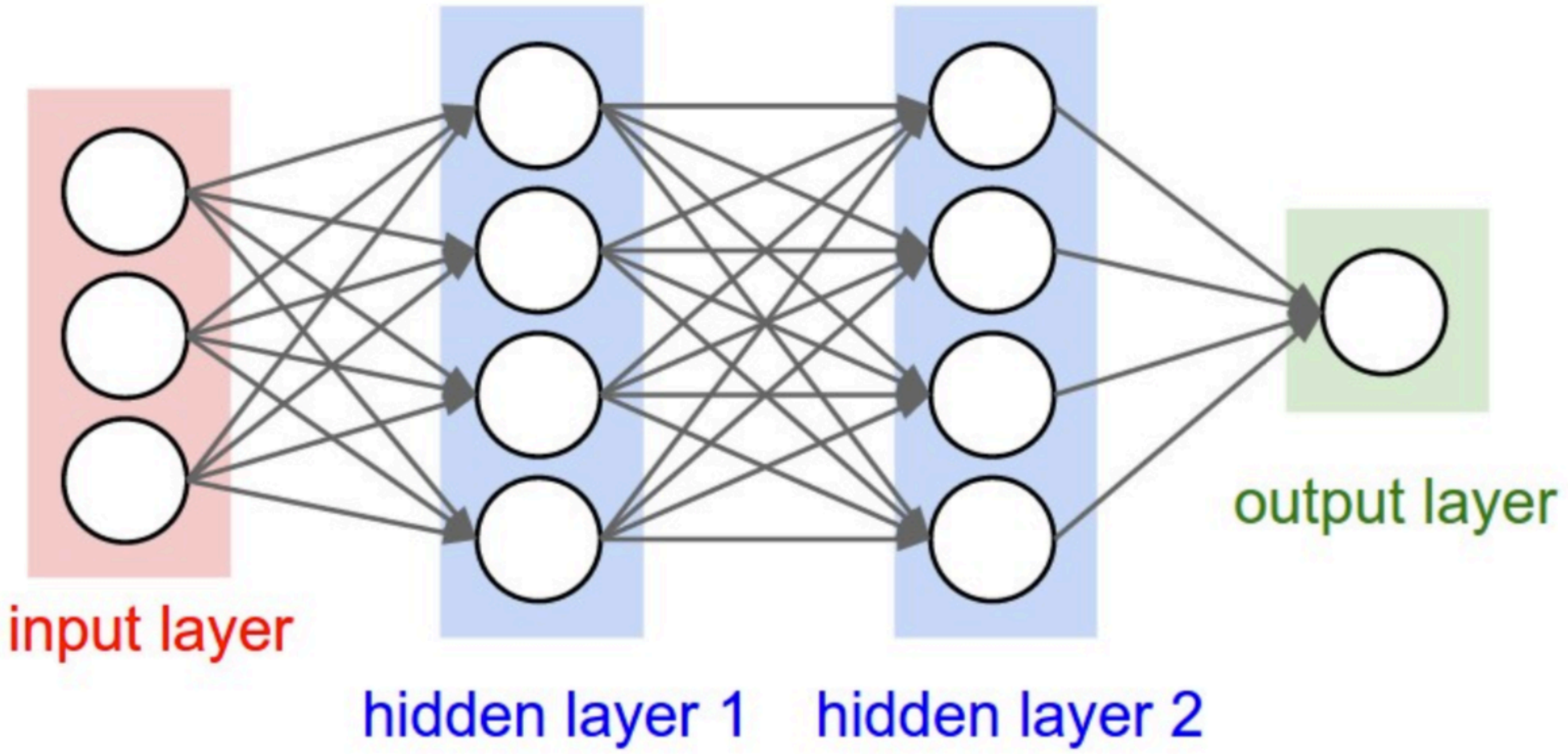




$$f(z) = \frac{1}{1 + e^{-z}}$$

$$h_{\mathbf{w},b}(\mathbf{x}) = f(\mathbf{w}^\intercal \mathbf{x} + b)$$

A neural network
= running several logistic regressions at the same time



- If we feed a vector of inputs through a bunch of logistic regression functions, then we get a vector of outputs…
- which we can feed into another logistic regression function

A closer look at activation functions and expressiveness of neural networks

Why non-linearities?

- Neural networks can learn much more complex functions and nonlinear decision boundaries

The capacity of the network increases with more hidden units and more hidden layers

What if we remove activation function?

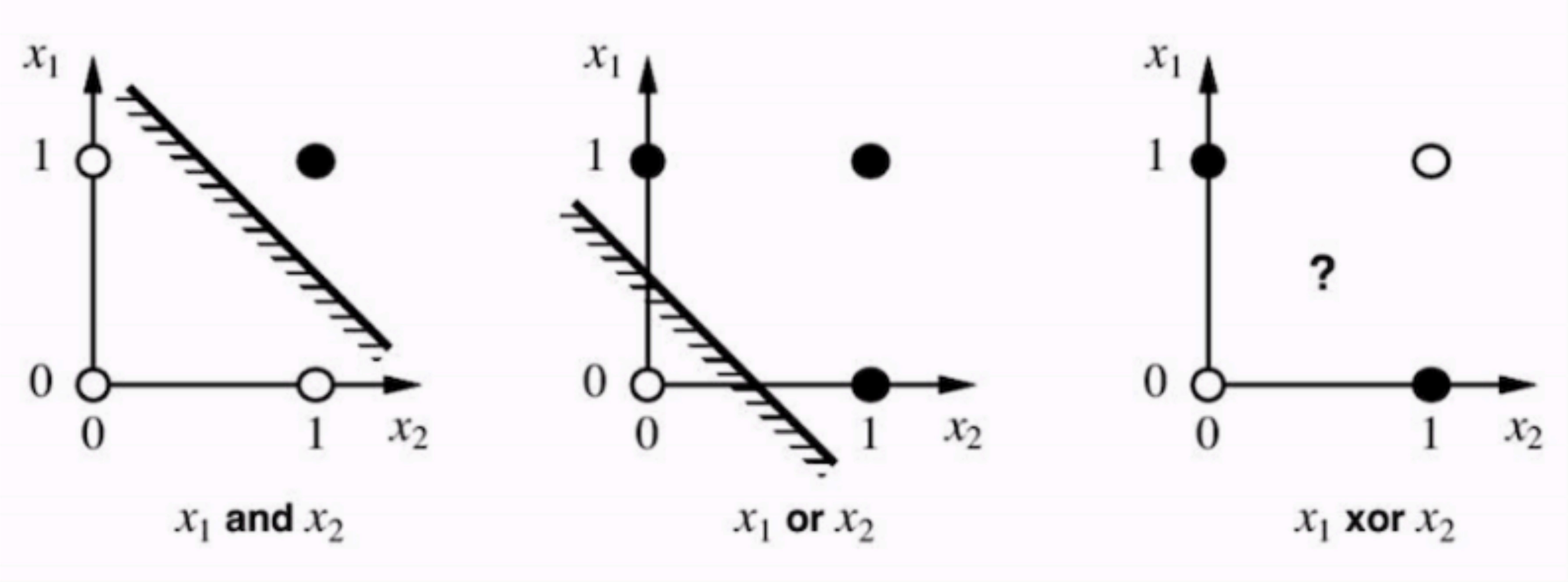
XOR problem

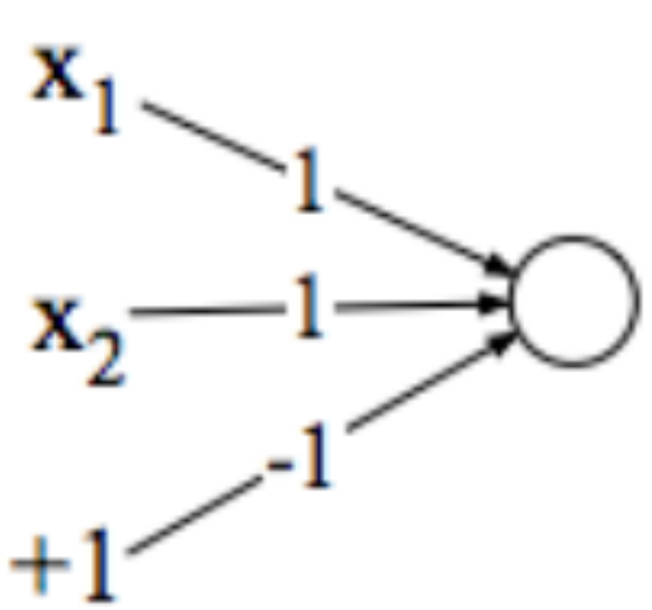
We will assume a training set where each label is in the set $\mathcal{Y} = \{-1, +1\}$

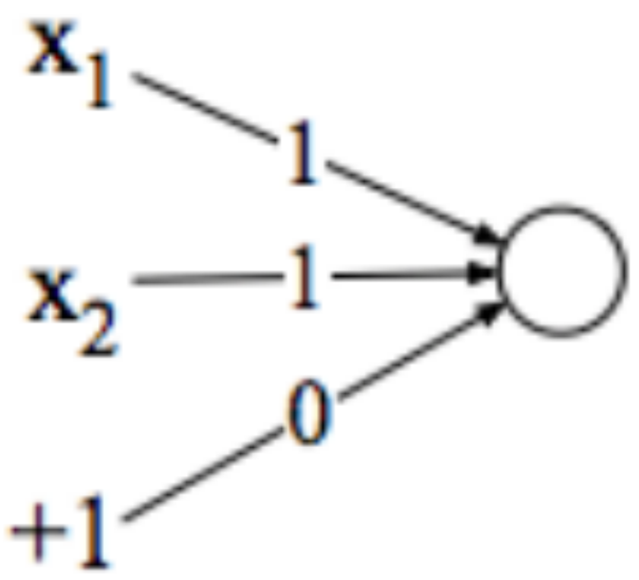
There are four training examples:

$$
\begin{aligned}
x^1 &= [0,0], y^1 = -1 \\
x^2 &= [0,1], y^2 = +1 \\
x^3 &= [1,0], y^3 = +1 \\
x^4 &= [1,1], y^4 = -1
\end{aligned}
$$

Single neuron (perceptron)







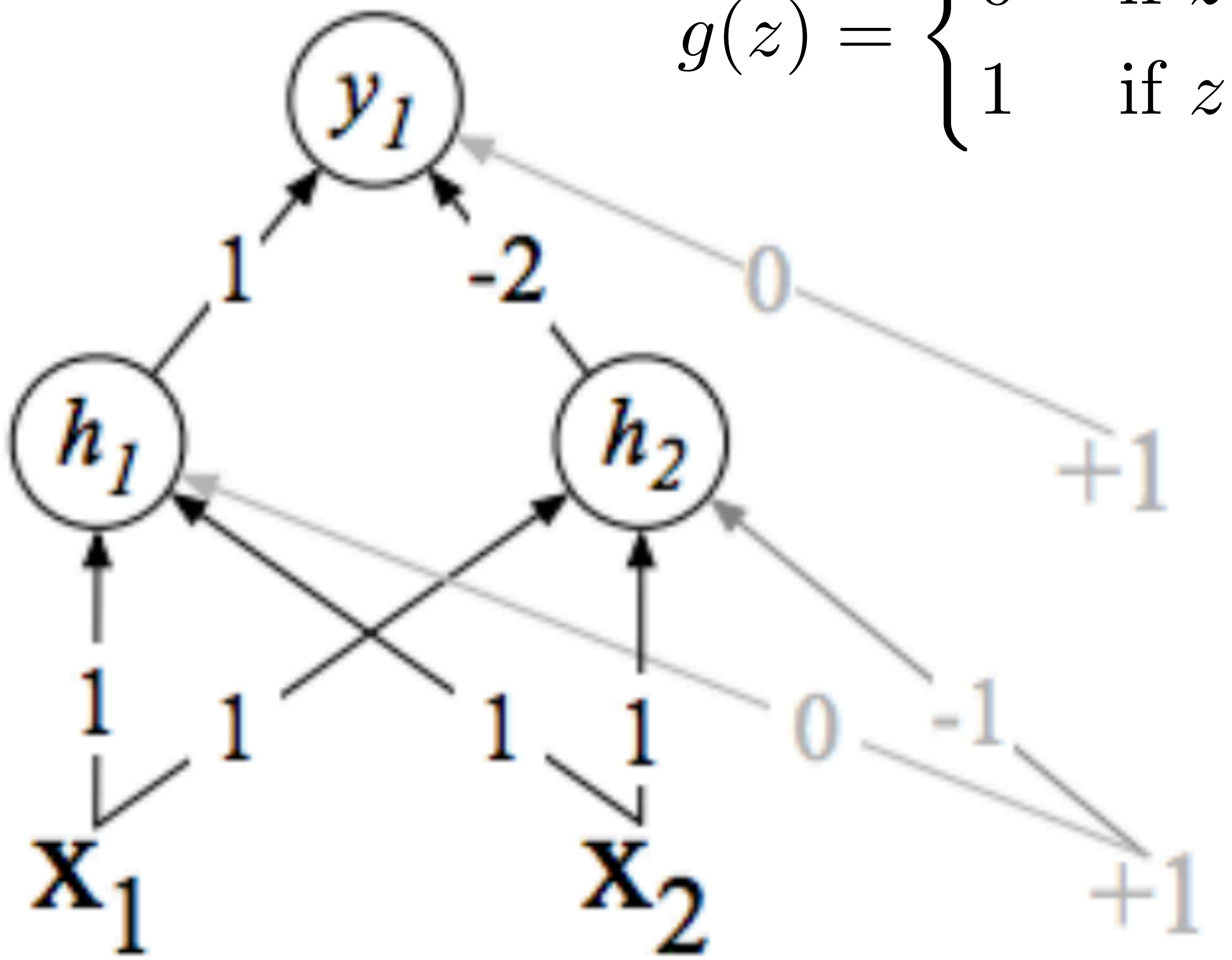
Perceptron can compute **and** and **or**

$$y = \begin{cases} 0 & \text{if } \mathbf{w} \cdot \mathbf{x} + b \leq 0 \\ 1 & \text{if } \mathbf{w} \cdot \mathbf{x} + b > 0 \end{cases}$$

Multiple neurons for XOR

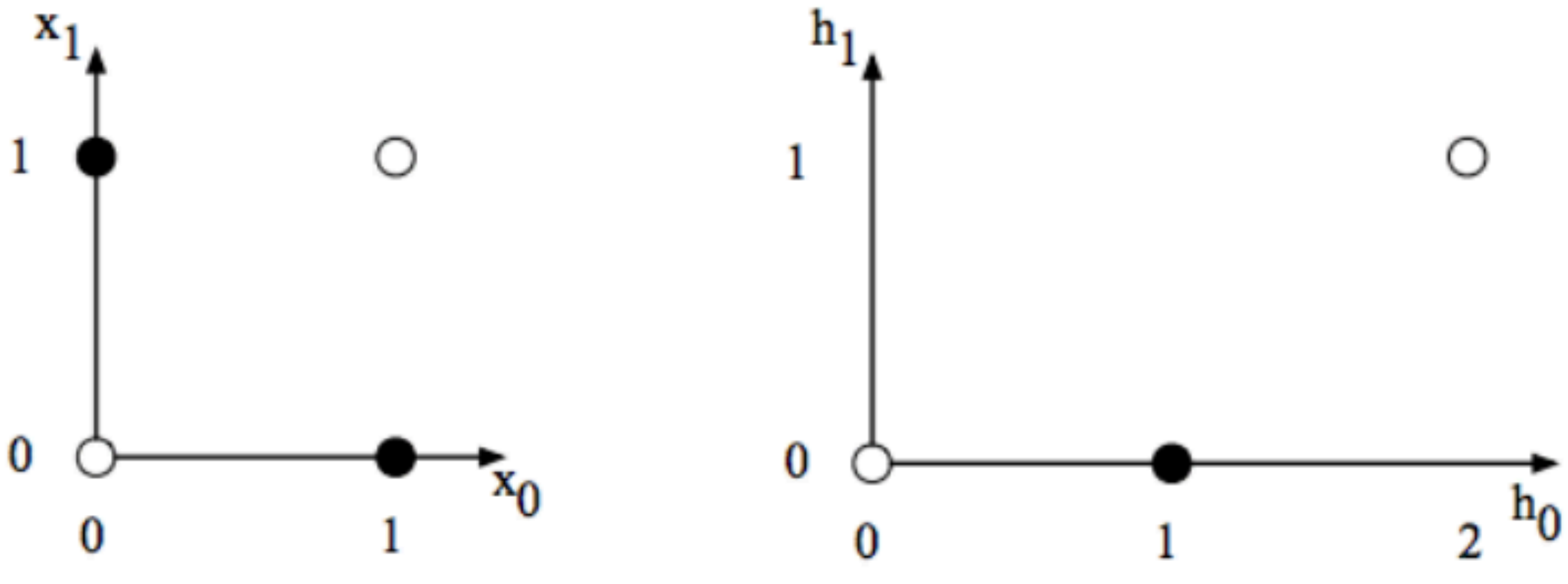
$$y_1 = g(h_1 - 2h_2)$$

$$g(z) = \begin{cases} 0 & \text{if } z \leq 0 \\ 1 & \text{if } z > 0 \end{cases}$$



$$h_1 = g(x_1 + x_2)$$

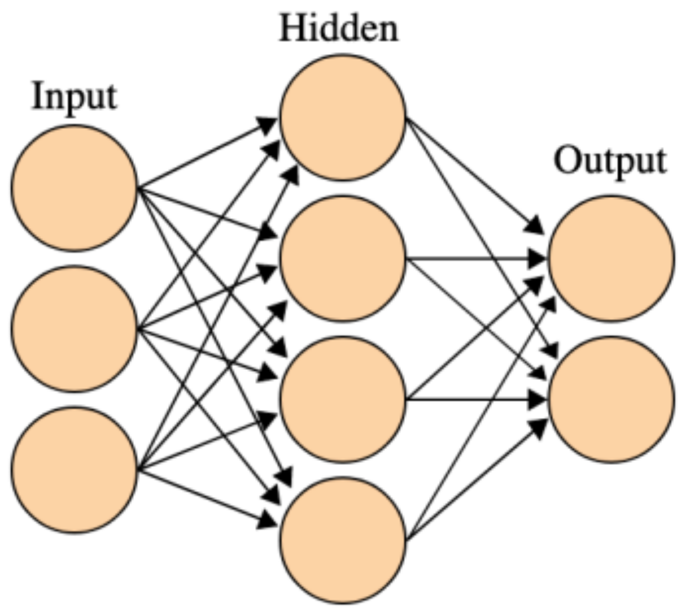
$$h_2 = g(x_1 + x_2 - 1)$$



a) The original x space

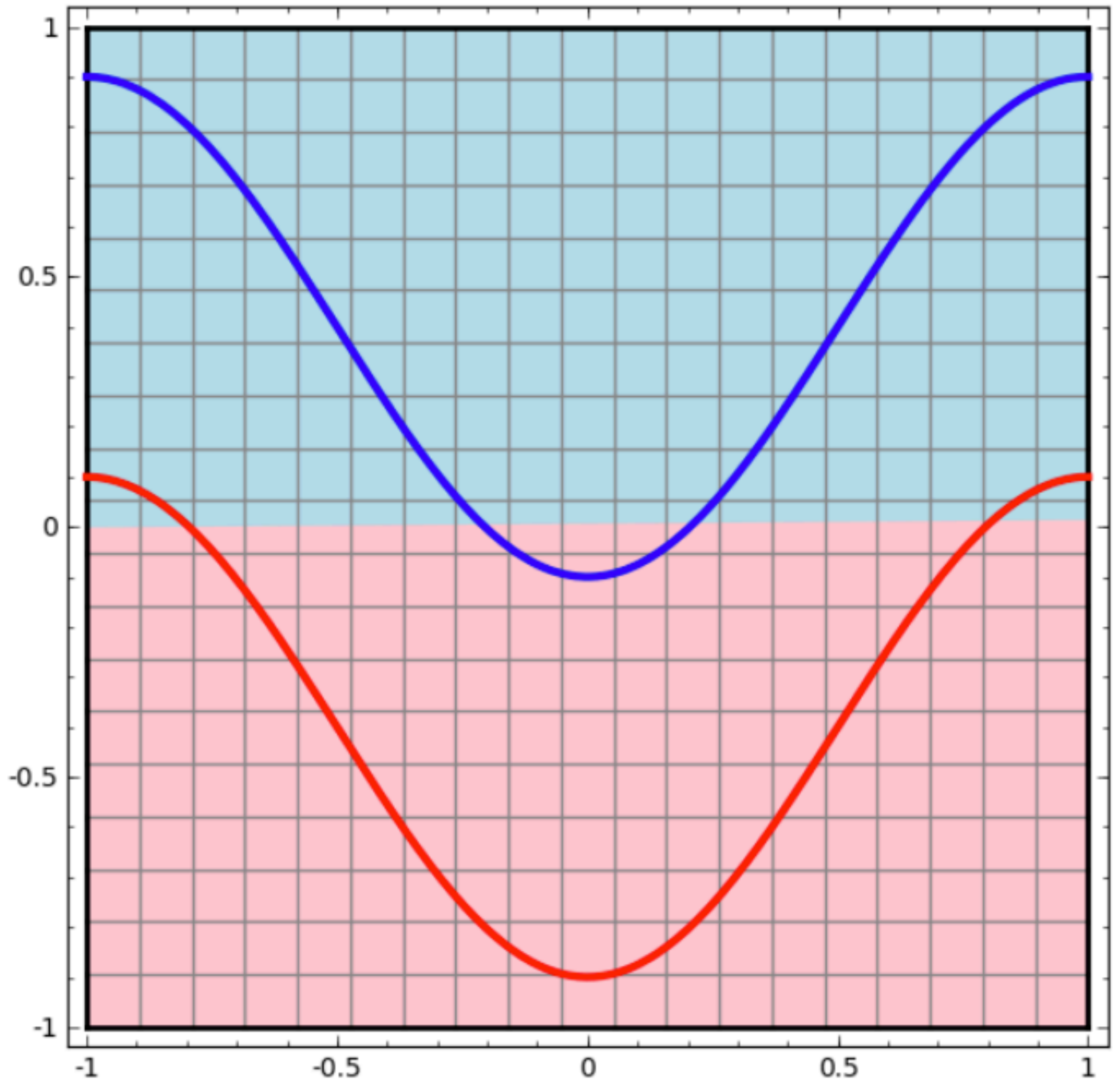
b) The new h space

Why nonlinearities



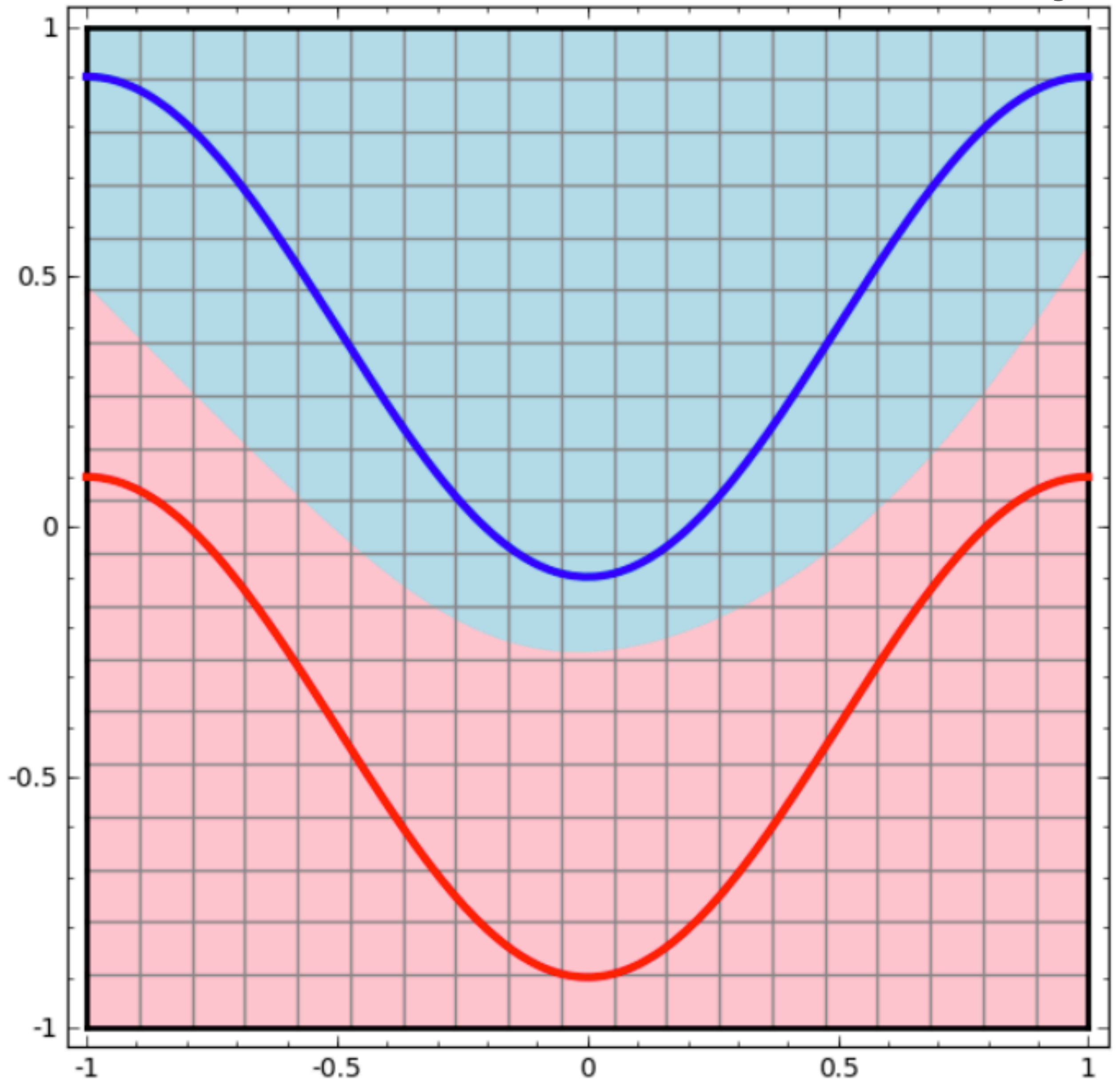
Learn to classify whether points should belong to the blue curve or red curve

Linear decision boundary



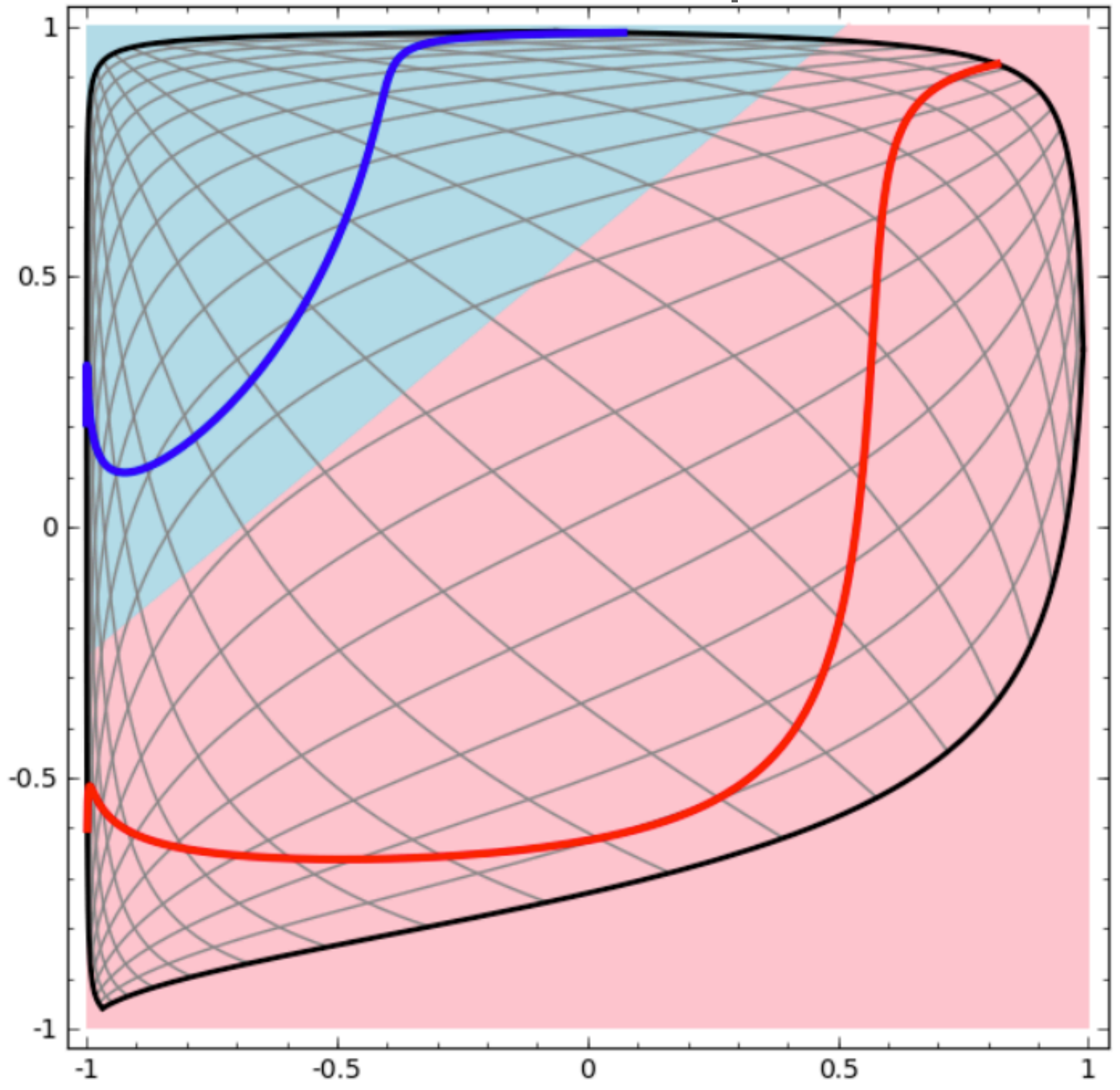
Linear classifier

Non-linear decision boundary



Neural Networks

Linear decision boundary
in transformed space



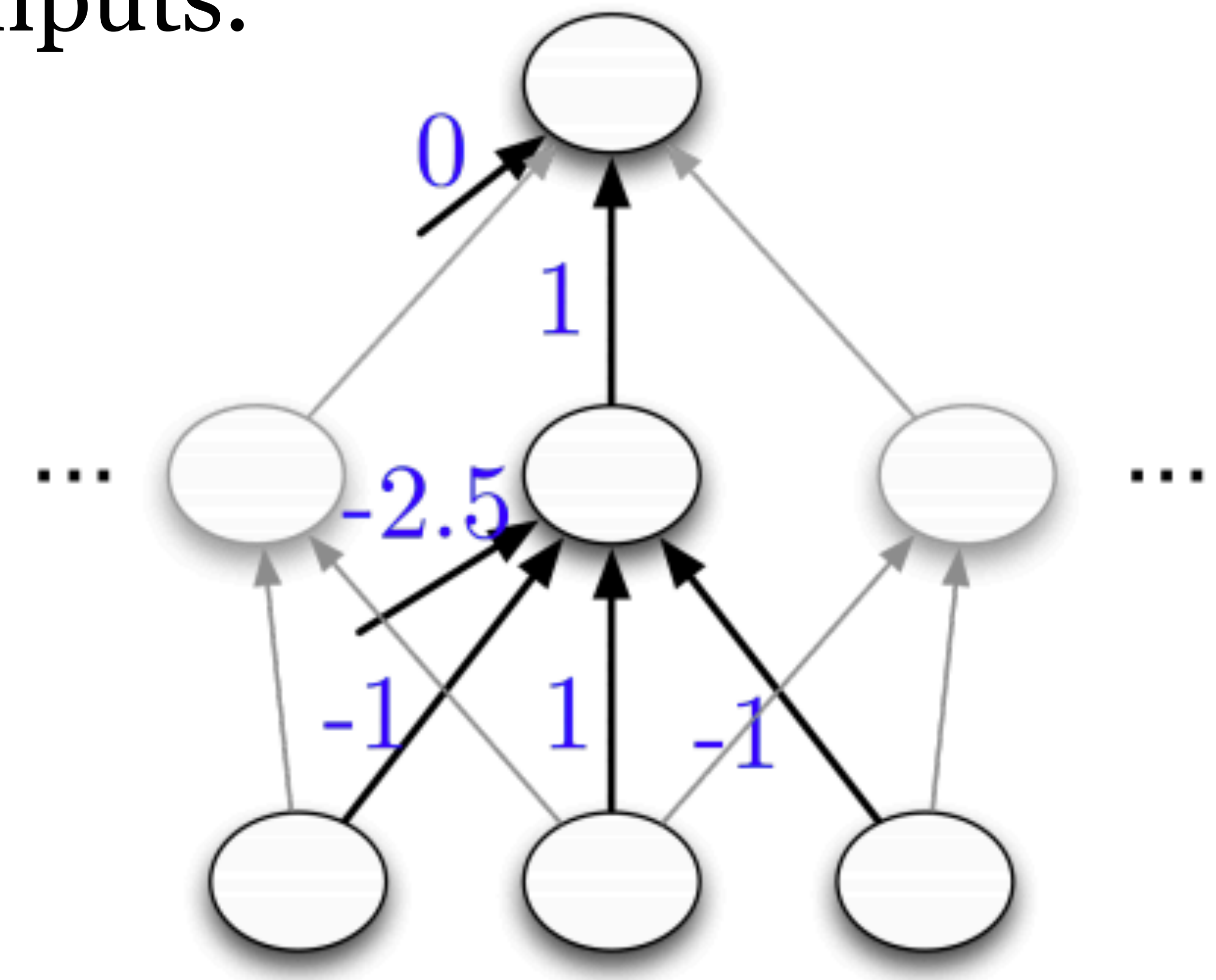
Hidden representations
are linearly separable!

https://colah.github.io/posts/2014-03-NN-Manifolds-Topology/

Expressiveness of neural networks

- Multilayer feed-forward neural nets with nonlinear activation functions are universal approximators
- True for both shallow networks (infinitely wide) and (infinitely) deep networks.
- Consider a network with just 1 hidden layer (with hard threshold activation functions) and a linear output. By having 2^D hidden units, each of which responds to just one input configuration, can model any boolean function with D inputs.

x_1	x_2	x_3	t
	$\vdots$		$\vdots$
-1	-1	1	-1
-1	1	-1	1
-1	1	1	1
	$\vdots$		$\vdots$

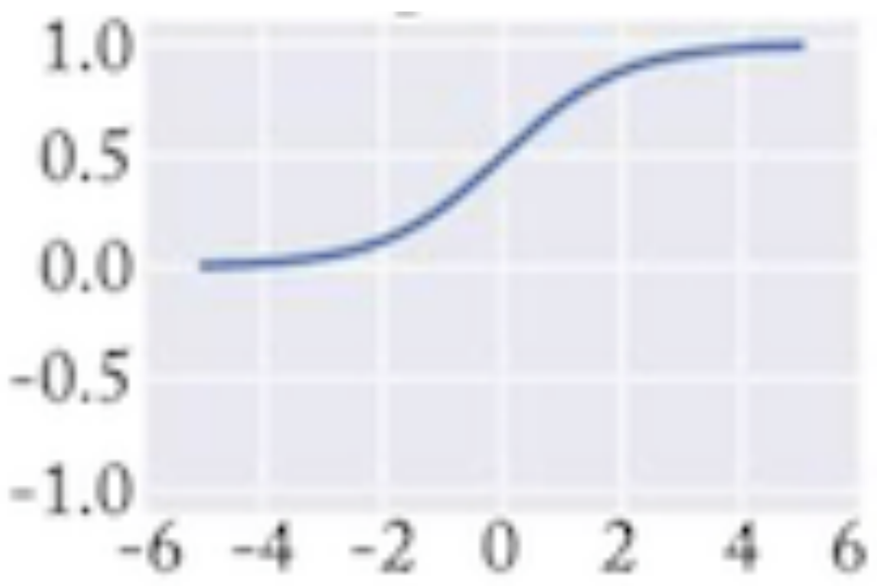


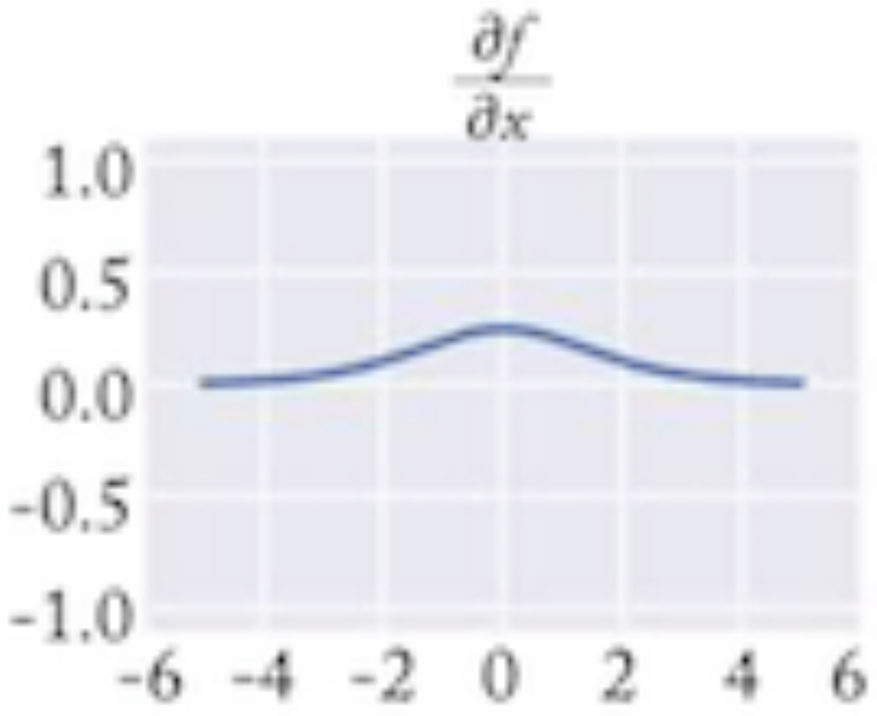
- Deep network can provide a more compact representation

Activation functions

sigmoid

$$f(z) = \frac{1}{1+e^{-z}}$$



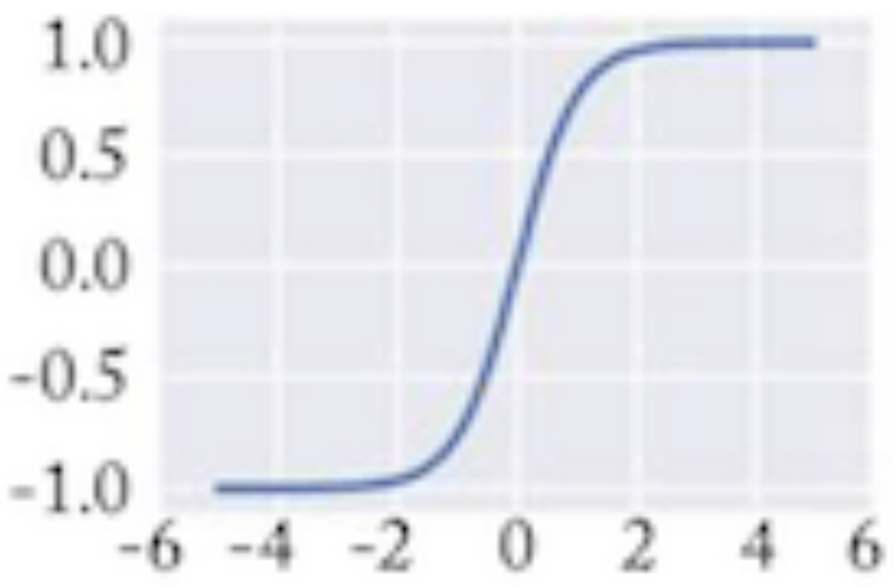


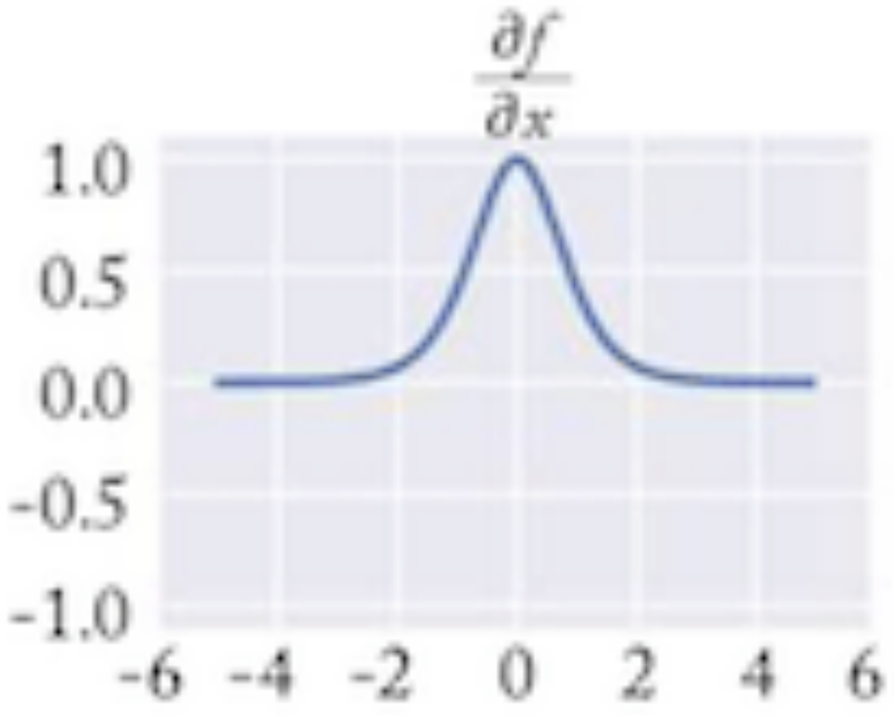
$$f'(z) = f(z) \times (1 - f(z))$$

Zero centered

tanh

$$f(z) = \frac{e^{2z}-1}{e^{2z}+1}$$



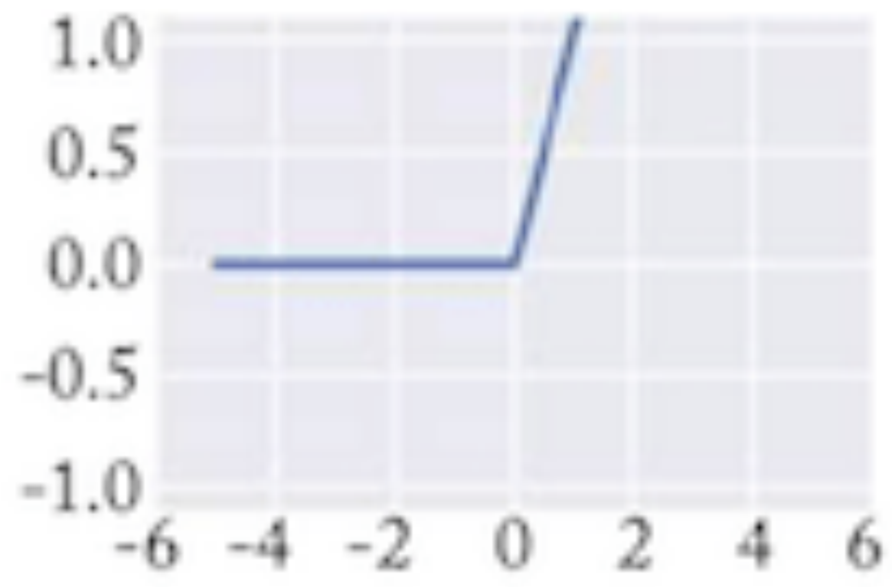


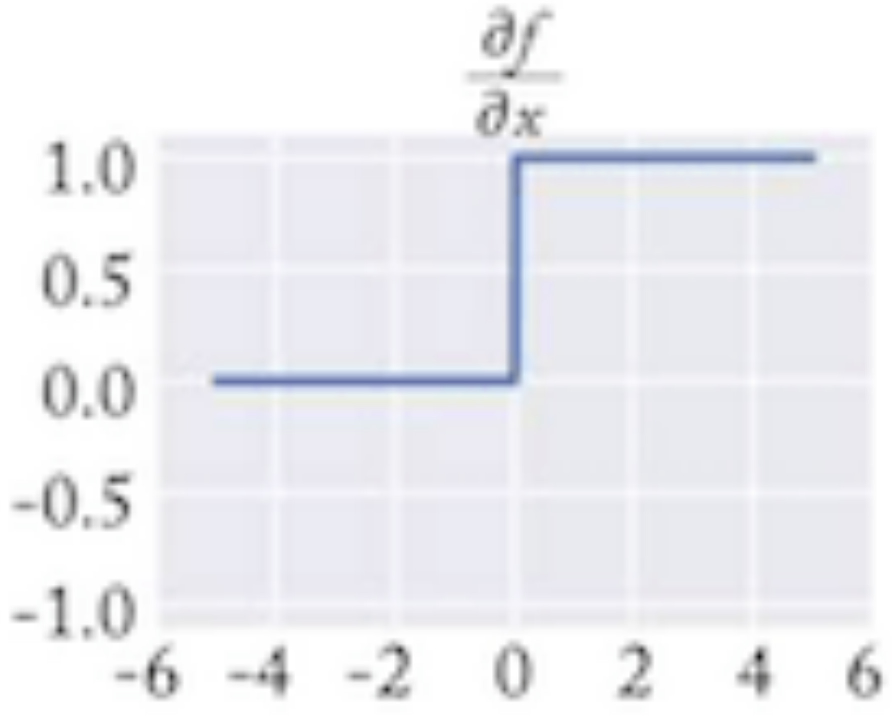
$$f'(z) = 1 - f(z)^2$$

Advantages of ReLU?

ReLU
(rectified linear unit)

$$f(z) = \max(0, z)$$





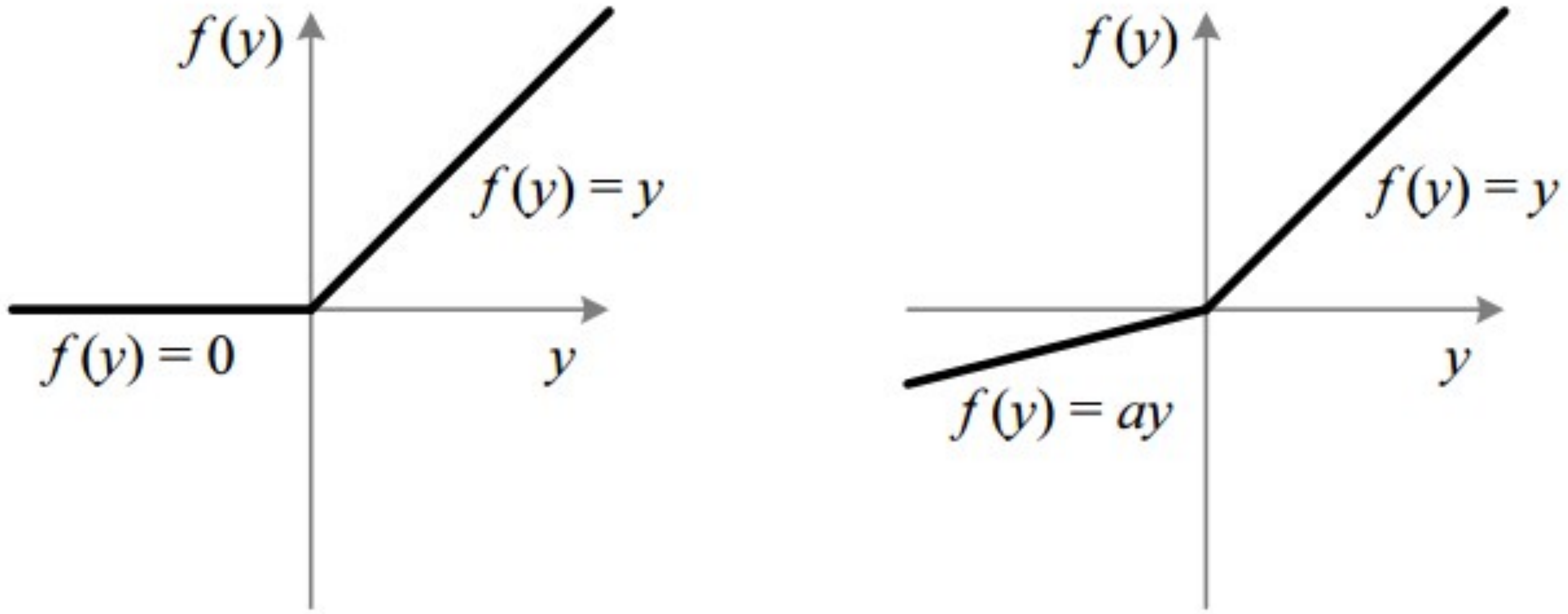
$$f'(z) = \begin{cases} 1 & z > 0 \\ 0 & z < 0 \end{cases}$$

Activation functions

Problems of ReLU? “dead neurons”

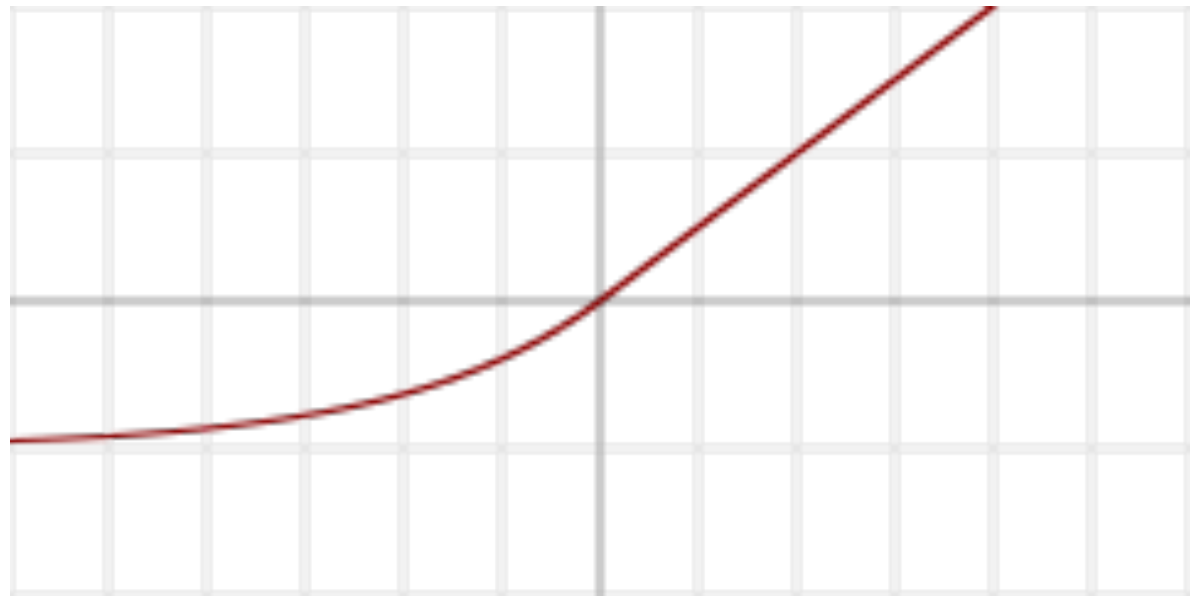
Leaky ReLU

$$f(z) = \begin{cases} z & z \geq 0 \\ 0.01z & z < 0 \end{cases}$$



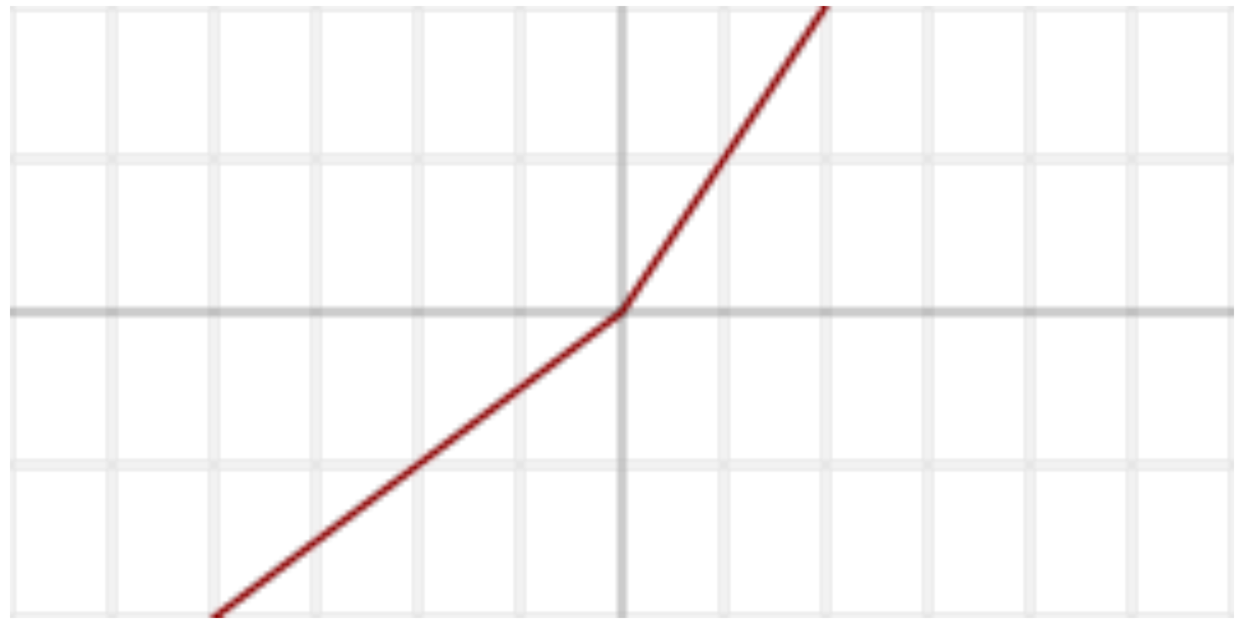
What is the best activation function?

- Depends on the problem!
- ReLU/Leaky ReLU is often a good choice
- Research into families of activation functions



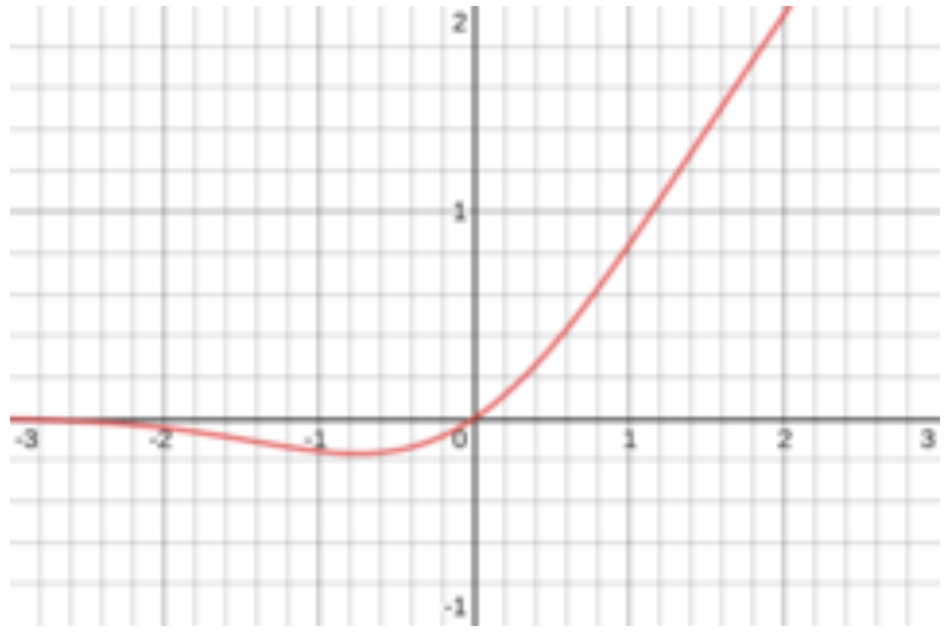
Exponential rectified linear unit (ELU)

$$f(\alpha, x) = \begin{cases} \alpha(e^x - 1) & \text{for } x \leq 0 \\ x & \text{for } x > 0 \end{cases}$$

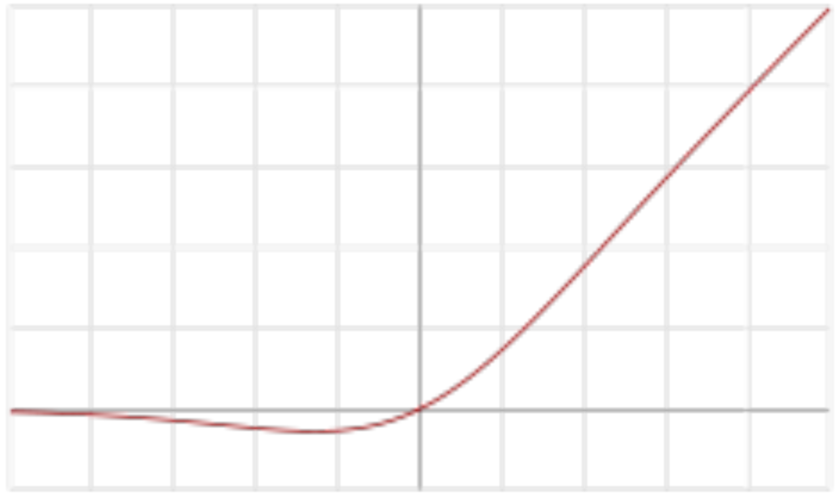


Parameteric rectified linear unit (PReLU)

$$f(\alpha, x) = \begin{cases} \alpha x & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$



$$\frac{1}{2}x + (1 + \mathrm{erf}(\frac{x}{\sqrt{2}}))$$



Swish

$$f(x) = x\sigma(\beta x) = \frac{x}{1 + e^{-\beta x}}$$

$$\beta = 0 \rightarrow f(x) = x/2$$

$$\beta = \infty \rightarrow f(x) = \mathrm{ReLU}$$

α, β can either be constant or trainable parameter

Training neural networks

Mathematical Notations

- Input layer: $x_1, \ldots, x_d$
- Hidden layer 1: $h_1^{(1)}, h_2^{(1)}, \ldots, h_{d_1}^{(1)}$

$$h_1^{(1)} = f(W_{1,1}^{(1)} x_1 + W_{1,2}^{(1)} x_2 + \ldots + W_{1,d}^{(1)} x_d + b_1^{(1)})$$

$$h_2^{(1)} = f(W_{2,1}^{(1)} x_1 + W_{2,2}^{(1)} x_2 + \ldots + W_{2,d}^{(1)} x_d + b_2^{(1)})$$

$$\ldots$$

- Hidden layer 2: $h_1^{(2)}, h_2^{(2)}, \ldots, h_{d_2}^{(2)}$

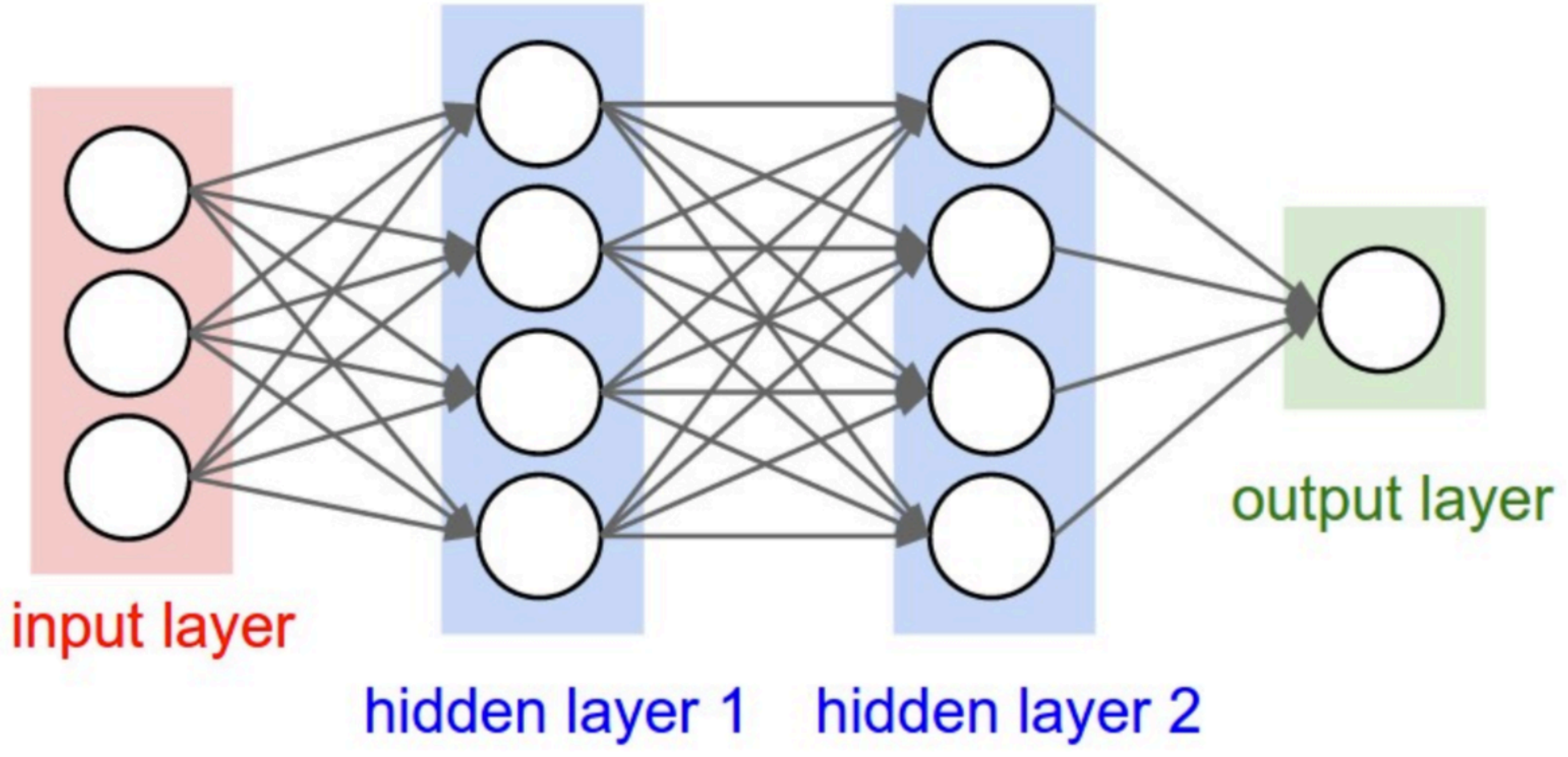
$$h_1^{(2)} = f(W_{1,1}^{(2)} h_1^{(1)} + W_{1,2}^{(2)} h_2^{(1)} + \ldots + W_{1,d_1}^{(2)} h_{d_1}^{(1)} + b_1^{(2)})$$

$$h_2^{(2)} = f(W_{2,1}^{(2)} h_1^{(1)} + W_{2,2}^{(2)} h_2^{(1)} + \ldots + W_{2,d_1}^{(2)} h_{d_1}^{(1)} + b_2^{(2)})$$

$$\ldots$$

- Output layer:

$$y = \sigma(w_1^{(o)} h_1^{(2)} + w_2^{(o)} h_2^{(2)} + \ldots + w_{d_2}^{(o)} h_{d_2}^{(2)} + b^{(o)})$$



Matrix Notations

- Input layer: $\mathbf{x} \in \mathbb{R}^d$
- Hidden layer 1:

$$\mathbf{h}_1 = f(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)}) \in \mathbb{R}^{d_1}$$

$$\mathbf{W}^{(1)} \in \mathbb{R}^{d_1 \times d}, \mathbf{b}^{(1)} \in \mathbb{R}^{d_1}$$

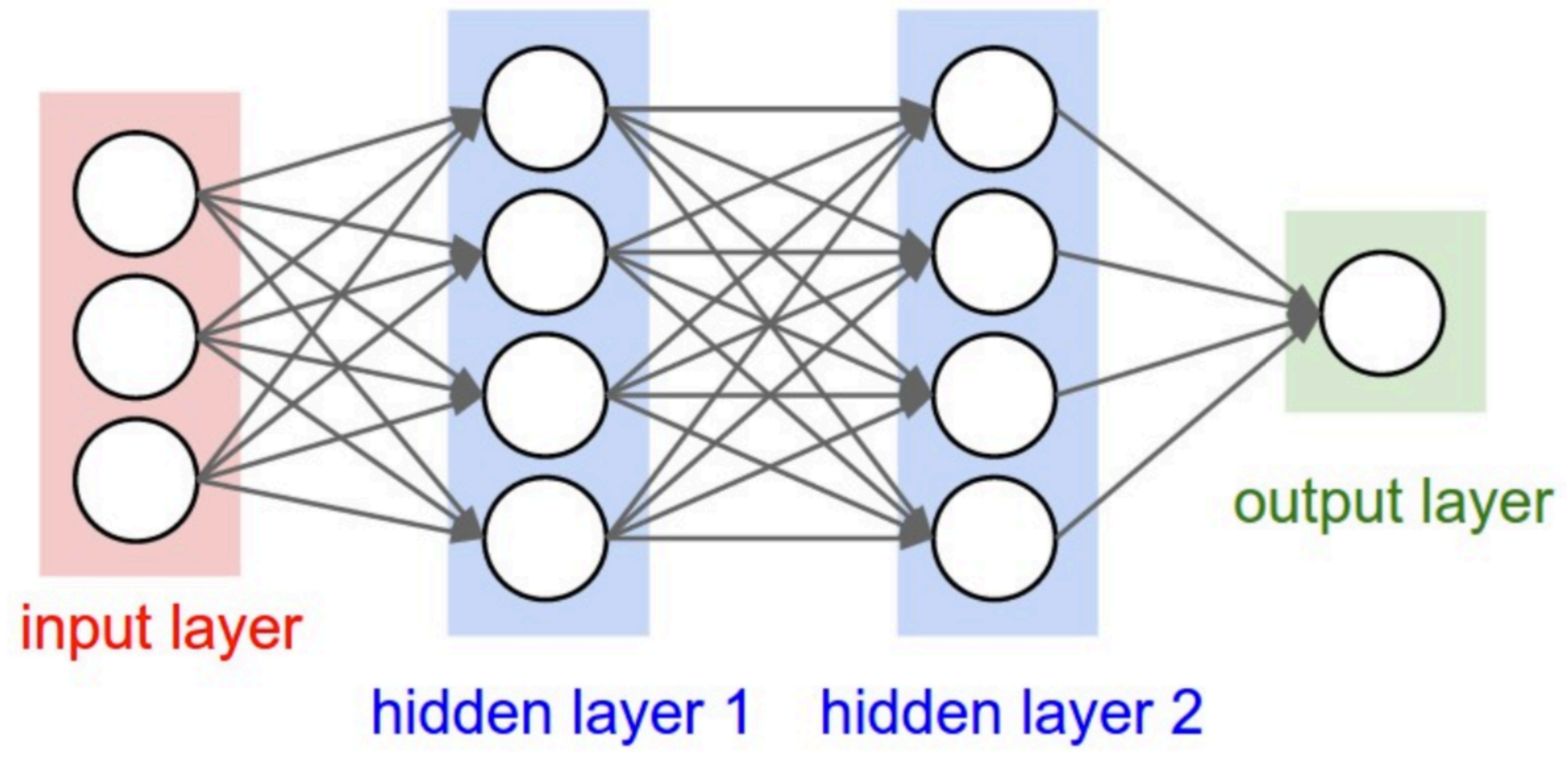
- Hidden layer 2:

$$\mathbf{h}_2 = f(\mathbf{W}^{(2)}\mathbf{h}_1 + \mathbf{b}^{(2)}) \in \mathbb{R}^{d_2}$$

$$\mathbf{W}^{(2)} \in \mathbb{R}^{d_2 \times d_1}, \mathbf{b}^{(2)} \in \mathbb{R}^{d_2}$$

- Output layer:

$$y = \sigma(\mathbf{w}^{(o)} \cdot \mathbf{h}_2 + b^{(o)})$$



Note: f is applied element-wise

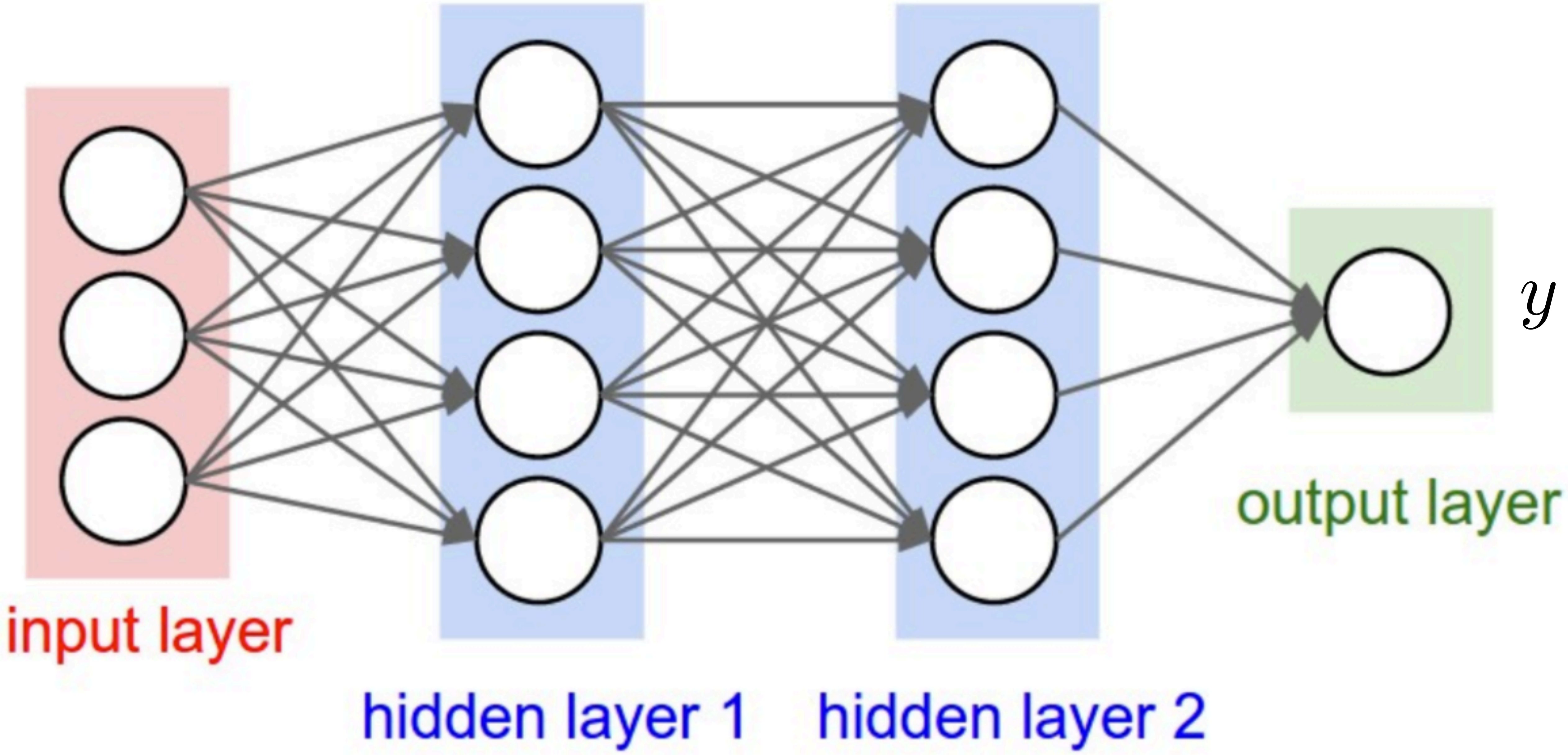
$$f([z_1, z_2, z_3]) = [f(z_1), f(z_2), f(z_3)]$$

Matrix Notation

- Learn parameters $\theta = \{\mathbf{W}^{(1)}, \mathbf{b}^{(1)}, \mathbf{W}^{(2)}, \mathbf{b}^{(2)}, \mathbf{w}^{(o)}, b^{(o)}\}$

Input

$\mathbf{x} \in \mathbb{R}^d$



Output

$y = \sigma(\mathbf{w}^{(o)\intercal}\mathbf{h}^{(o)} + b^{(o)})$

$\mathbf{w}^{(o)} \in \mathbb{R}^{d_2}$

$b^{(o)} \in \mathbb{R}$

$\mathbf{h}^{(1)} = \tanh(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$ $\quad$ $\mathbf{h}^{(2)} = \tanh(\mathbf{W}^{(2)}\mathbf{h}^{(1)} + \mathbf{b}^{(2)})$

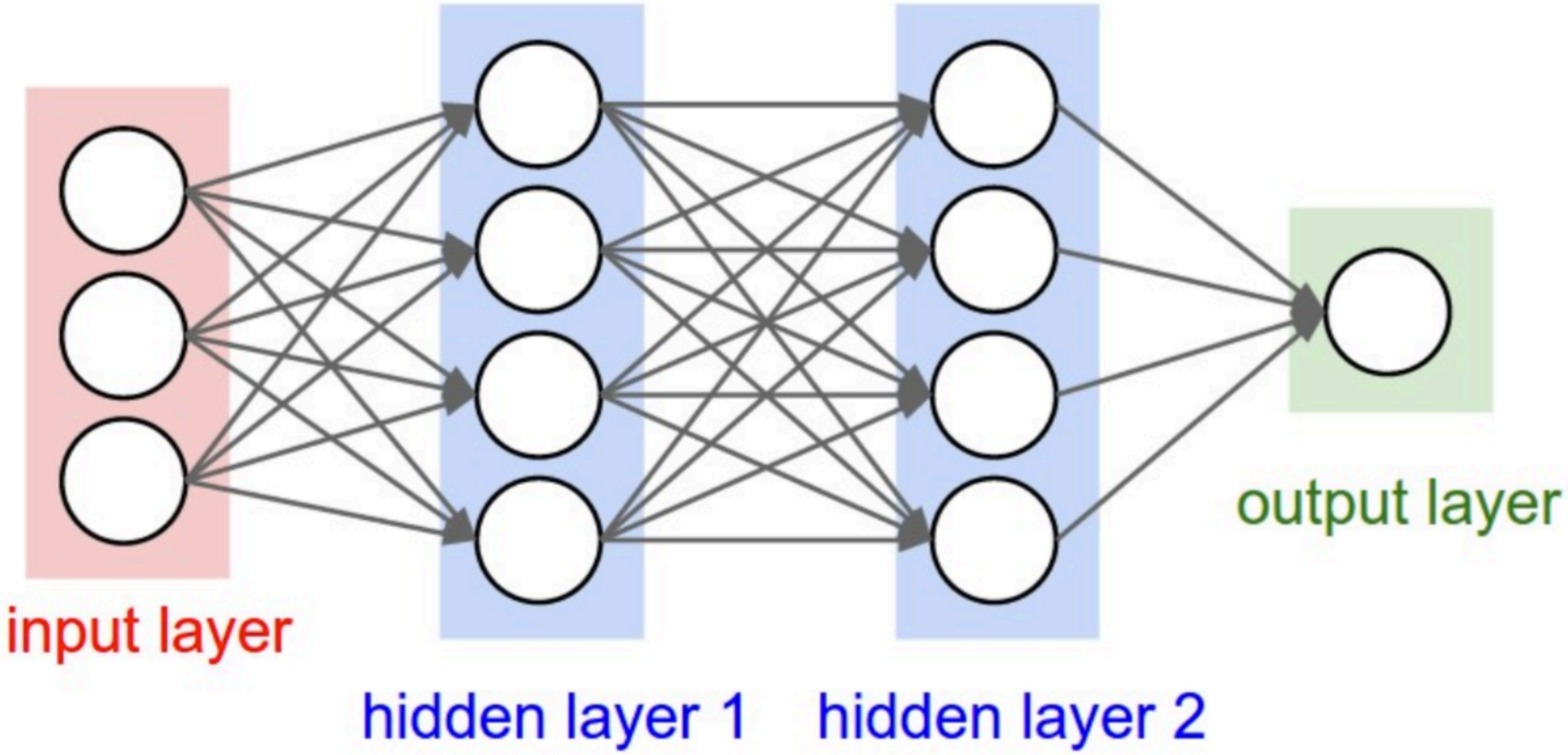
$\mathbf{W}^{(1)} \in \mathbb{R}^{d_1 \times d}$ $\quad$ $\mathbf{b}^{(1)} \in \mathbb{R}^{d_1}$ $\quad$ $\mathbf{W}^{(2)} \in \mathbb{R}^{d_2 \times d_1}$ $\quad$ $\mathbf{b}^{(2)} \in \mathbb{R}^{d_2}$

Matrix Notation

- Learn parameters $\theta = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{w}, b\}$

Input

$\mathbf{x} \in \mathbb{R}^d$



Output

$y = \sigma(\mathbf{w}^\intercal \mathbf{h_2} + b)$

$\mathbf{w} \in \mathbb{R}^{d_2}$

$\mathbf{h}_1 = \tanh(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1)$ $\qquad$ $\mathbf{h}_2 = \tanh(\mathbf{W}_2 \mathbf{h}_1 + \mathbf{b}_2)$

$\mathbf{W}_1 \in \mathbb{R}^{d_1 \times d}$ $\quad$ $\mathbf{b}_1 \in \mathbb{R}^{d_1}$ $\quad$ $\mathbf{W}_2 \in \mathbb{R}^{d_2 \times d_1}$ $\quad$ $\mathbf{b}_2 \in \mathbb{R}^{d_2}$

Loss functions

- Binary classification

$$y = \sigma(\mathbf{w} \cdot \mathbf{h}_2 + b)$$

Ground truth label

$$\mathcal{L}(y, y^*) = -y^* \log y - (1 - y^*) \log (1 - y)$$

- Regression

$$y = \mathbf{w} \cdot \mathbf{h}_2 + b$$

$$\mathcal{L}_{\mathrm{MSE}}(y, y^*) = (y - y^*)^2$$

- Multi-class classification (C classes)

$$y_i = \mathrm{softmax}_i(\mathbf{W}\mathbf{h}_2 + \mathbf{b}) \qquad \mathbf{W} \in \mathbb{R}^{C \times d_2}, \mathbf{b} \in \mathbb{R}^C$$

$$\mathcal{L}(y, y^*) = -\sum_{i=1}^{C} y_i^* \log y_i$$

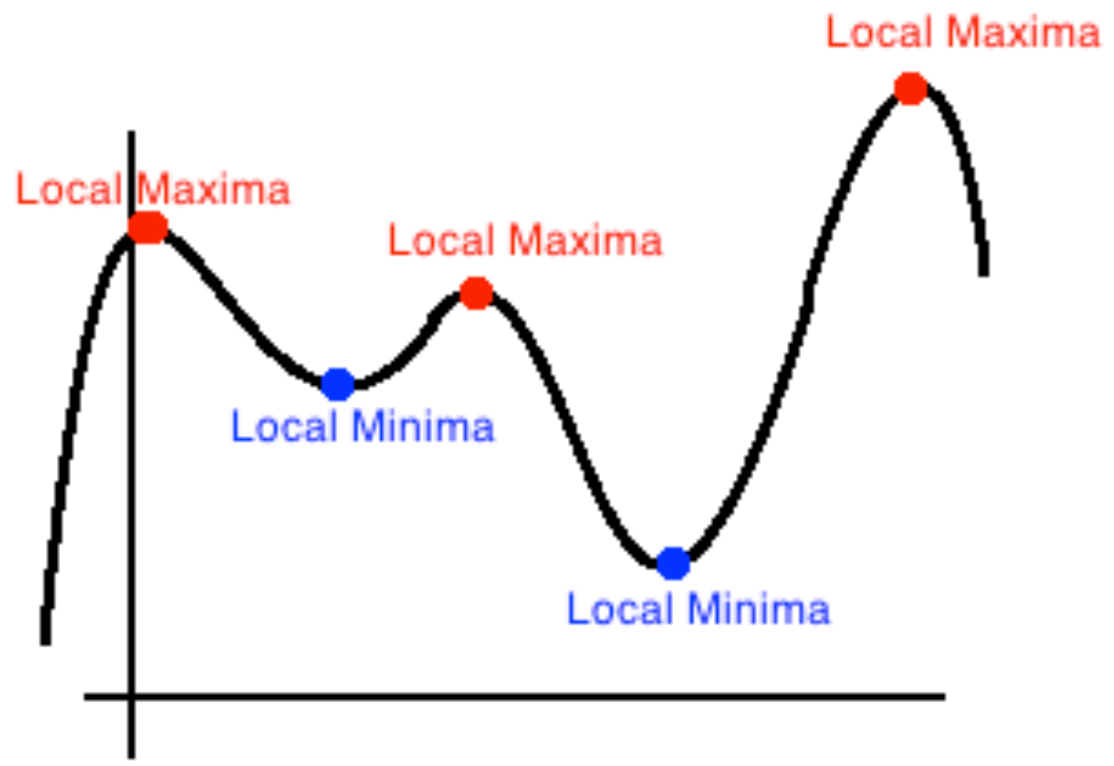
How find parameters to minimize $\mathcal{L}(\theta)$

$$\theta = \{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{w}, b\}$$

Optimization

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla_\theta J(\theta)$$

- Logistic regression is convex: one global minimum
- Neural networks are non-convex and not easy to optimize
- A class of more sophisticated "adaptive" optimizers that scale the parameter adjustment by an accumulated gradient.
  - **Adam**
  - Adagrad
  - RMSprop
  - ...



(Ruder 2016): An overview of gradient descent optimization algorithms (https://ruder.io/optimizing-gradient-descent/)

Standard SGD

- Simple SGD: update with only gradients

$$g_t = \nabla_{\theta_{t-1}} L(\theta_{t-1})$$ Gradient of Loss

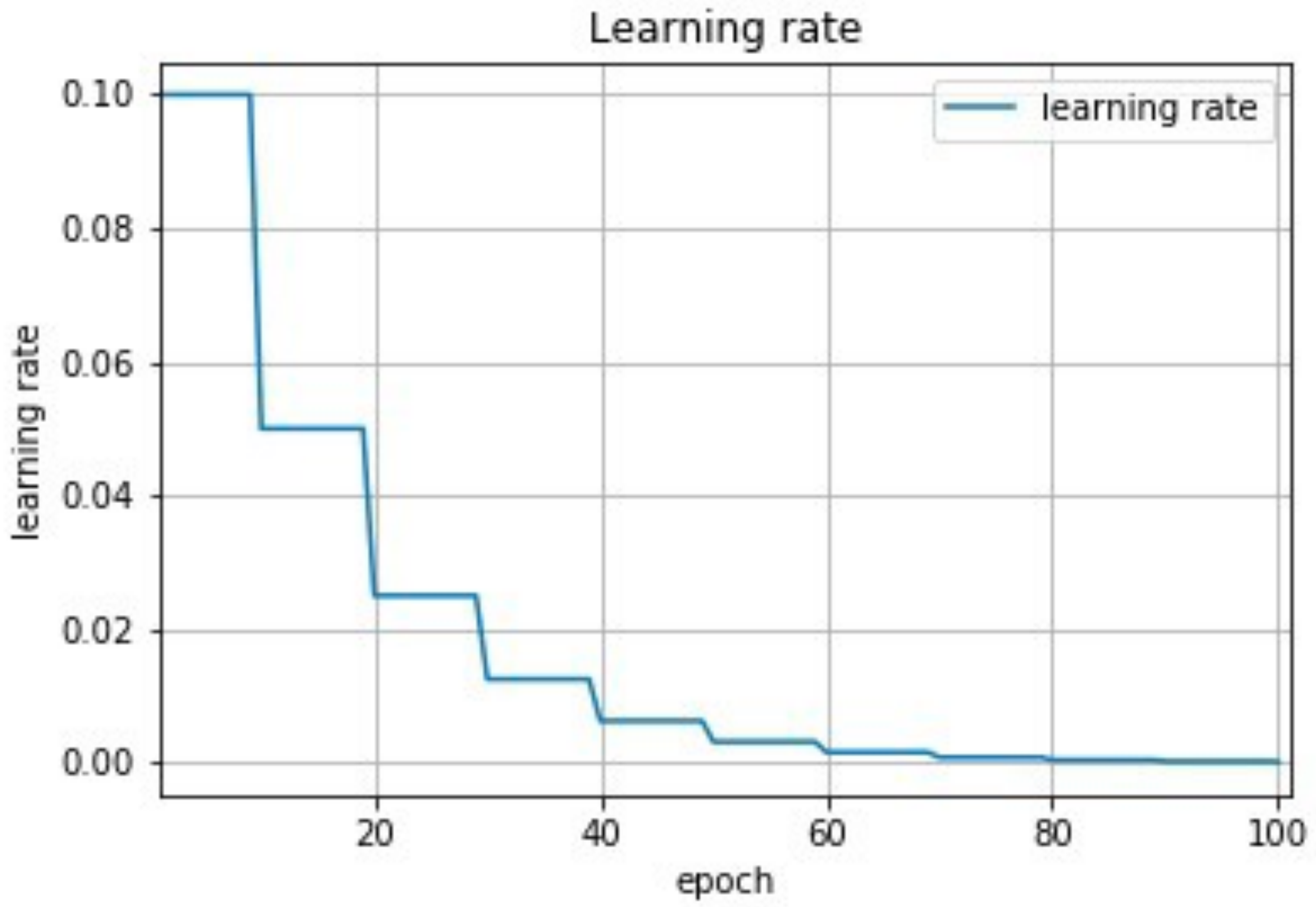
$$\theta_t = \theta_{t-1} - \underline{\eta} g_t$$

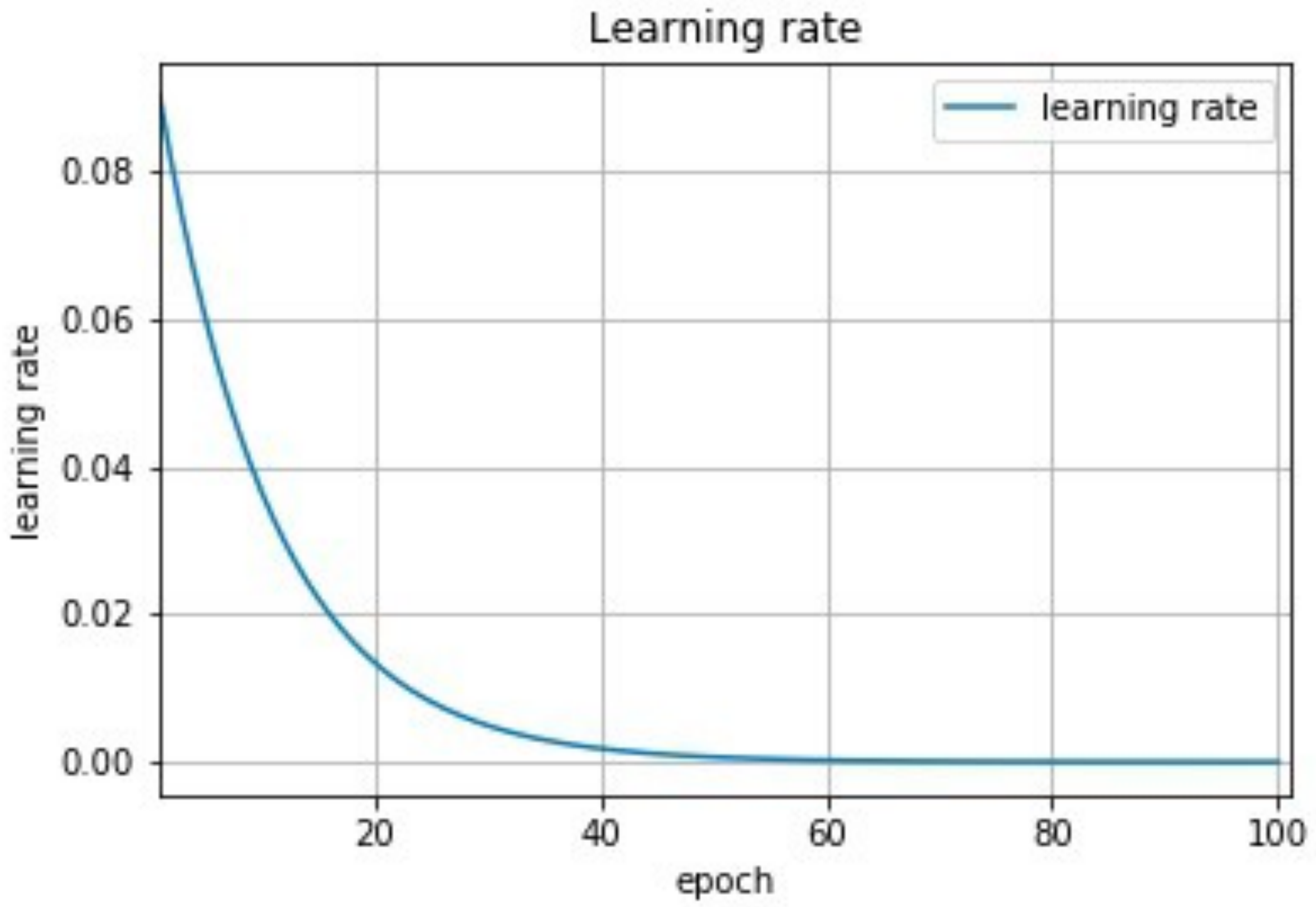
Learning Rate

Credits: Graham Neubig

Using SGD

- Decay learning rate





- Mini-batch (update after seeing m samples)
  - Less variance than pure SGD
  - More efficient than updating the weights after every sample
  - Randomize/shuffle samples in each mini-batch

There is a cost to updating weights

SGD with Momentum

- Remember gradients from past time steps

$$\underbrace{v_t}_{\text{Momentum}} = \underbrace{\gamma}_{\text{Momentum Conservation Parameter}} \underbrace{v_{t-1}}_{\text{Previous Momentum}} + \eta g_t$$

$$\theta_t = \theta_{t-1} - v_t$$

- Intuition: Prevent instability resulting from sudden changes

Adagrad

Don't have to fiddle with the learning rate parameter so much (use $\eta = 0.01$)

- Adaptively reduce learning rate based on accumulated variance of the gradients

$$G_t = G_{t-1} + g_t \odot g_t$$

Current Gradient Squared

Always positive

Grows larger and larger

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{G_t + \epsilon}} \odot g_t$$

Small constant

- Intuition: frequently updated parameters (e.g. common word embeddings) should be updated less

- Problem: learning rate continuously decreases, and training can stall (fixed by using rolling average in AdaDelta and RMSProp)

Credits: Graham Neubig

Adam

- Commonly used optimization option (for NLP and more)
- Considers rolling average of gradient and momentum

$$m_1 = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$ Momentum

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t \odot g_t$$ Rolling Average of Gradients

- Correction of bias early in training

$$\hat{m}_t = \frac{m_t}{1 - (\beta_1)^t} \qquad \hat{v}_t = \frac{v_t}{1 - (\beta_2)^t}$$

- Final update

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t$$

Optimization

- Standard/Vanilla SGD: update with only gradients
- Momentum: update w/ running average of gradients
  - Prevent instability from sudden changes
- Adagrad: different learning rate for each parameter
  - update down weighs high variance values
- Adam: update w/ running average of gradient, down weights by running average of variance

More details: TA tutorial on optimization

Blog: https://ruder.io/optimizing-gradient-descent/

Adapted from slide by Graham Neubig

Optimization

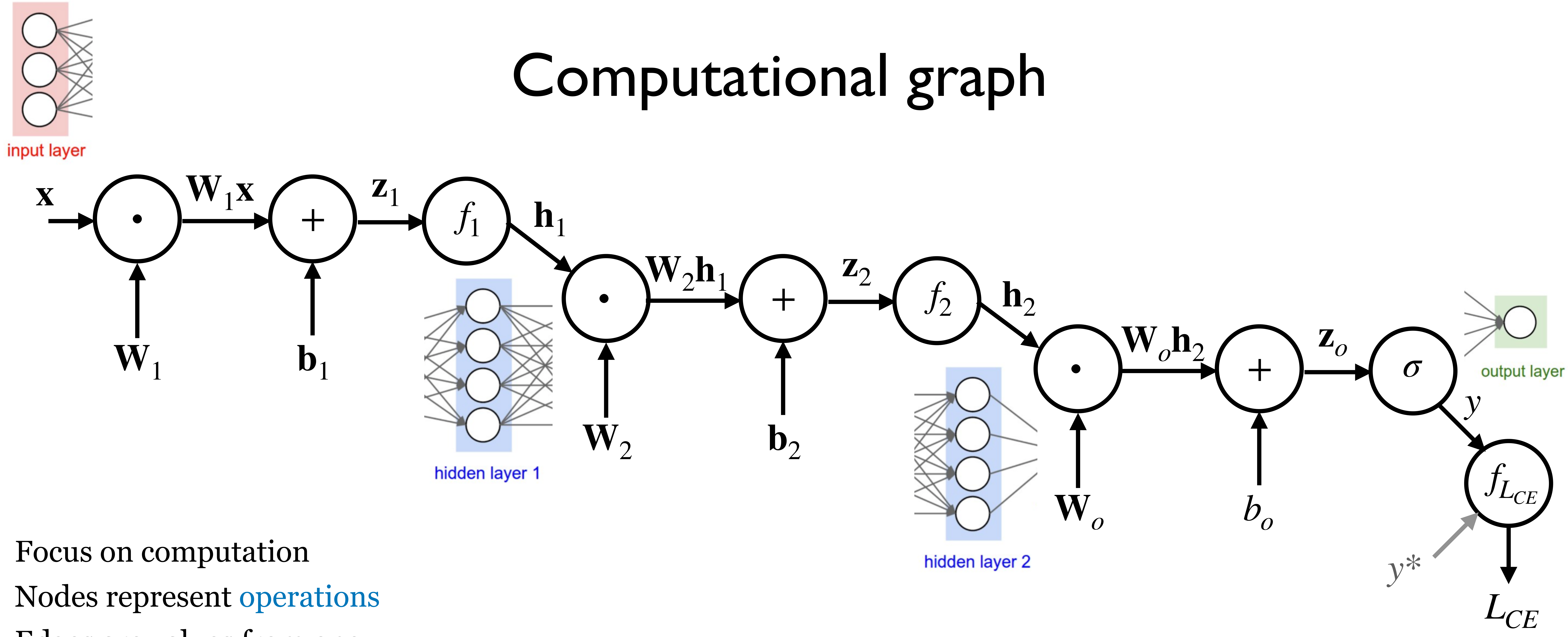
Gradient Descent

$$\theta^{(t+1)} = \theta^{(t)} - \eta \underline{\nabla_\theta \mathcal{L}(\theta)}$$

How to compute the gradient?

Neural networks as computational graphs

Computational graph



Focus on computation

Nodes represent operations

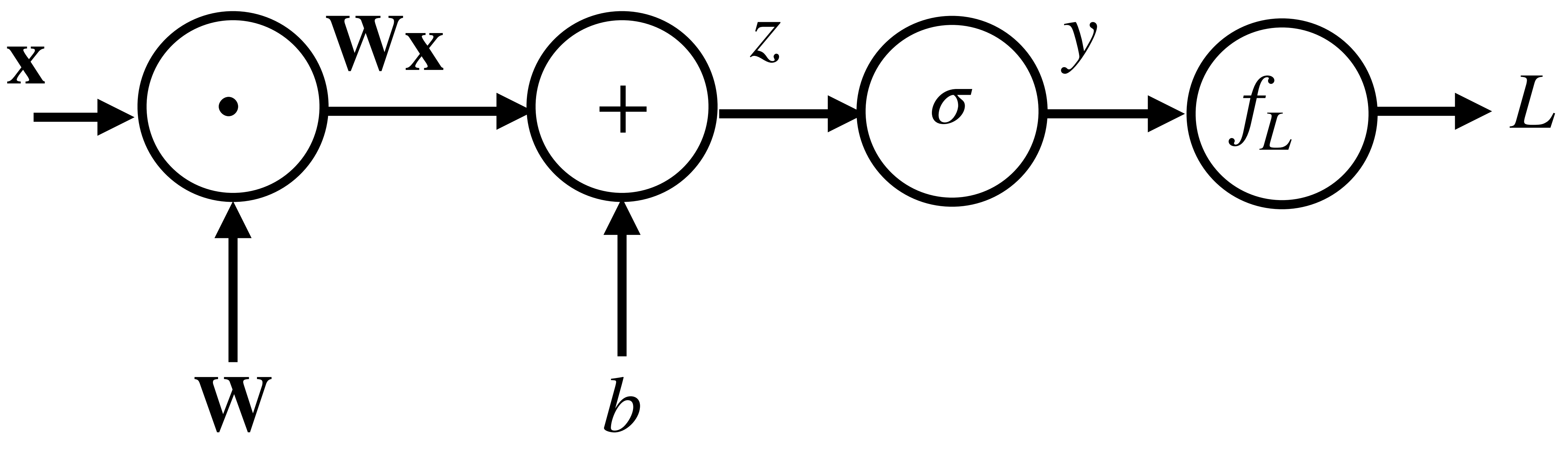
Edges are values from one operator to the next

Forward pass: compute function value

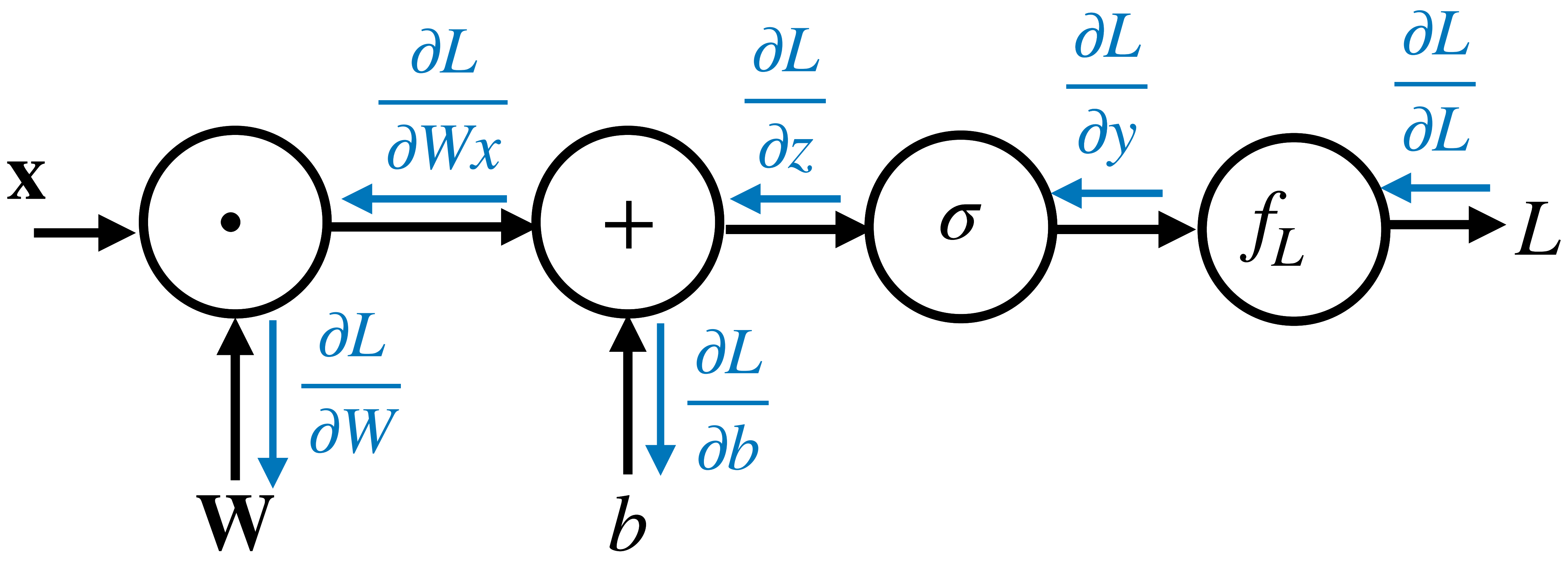
Backward pass: gradient using chain rule

Simplified example

Forward pass: compute function value



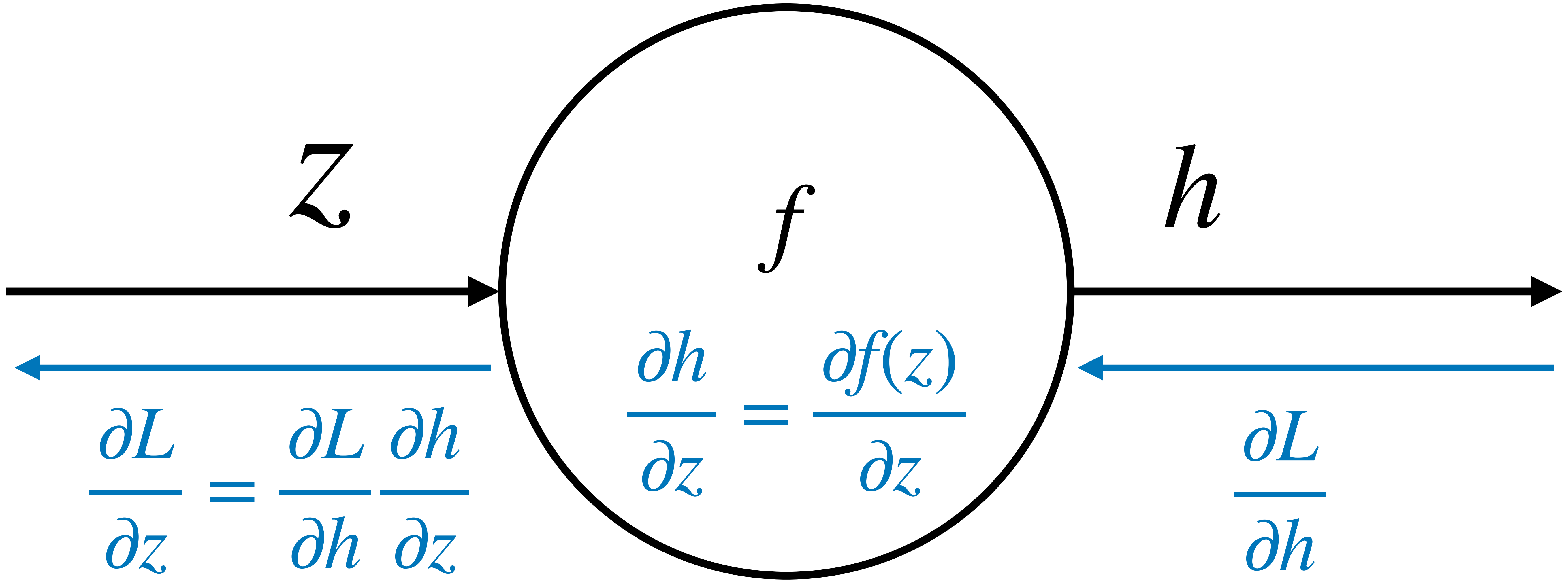
Backward pass: gradient using chain rule



- Good news is that modern automatic differentiation tools did all for you!
- Implementing backprop by hand is like programming in assembly language.

Backpropagation: single node

$$h = f(z)$$



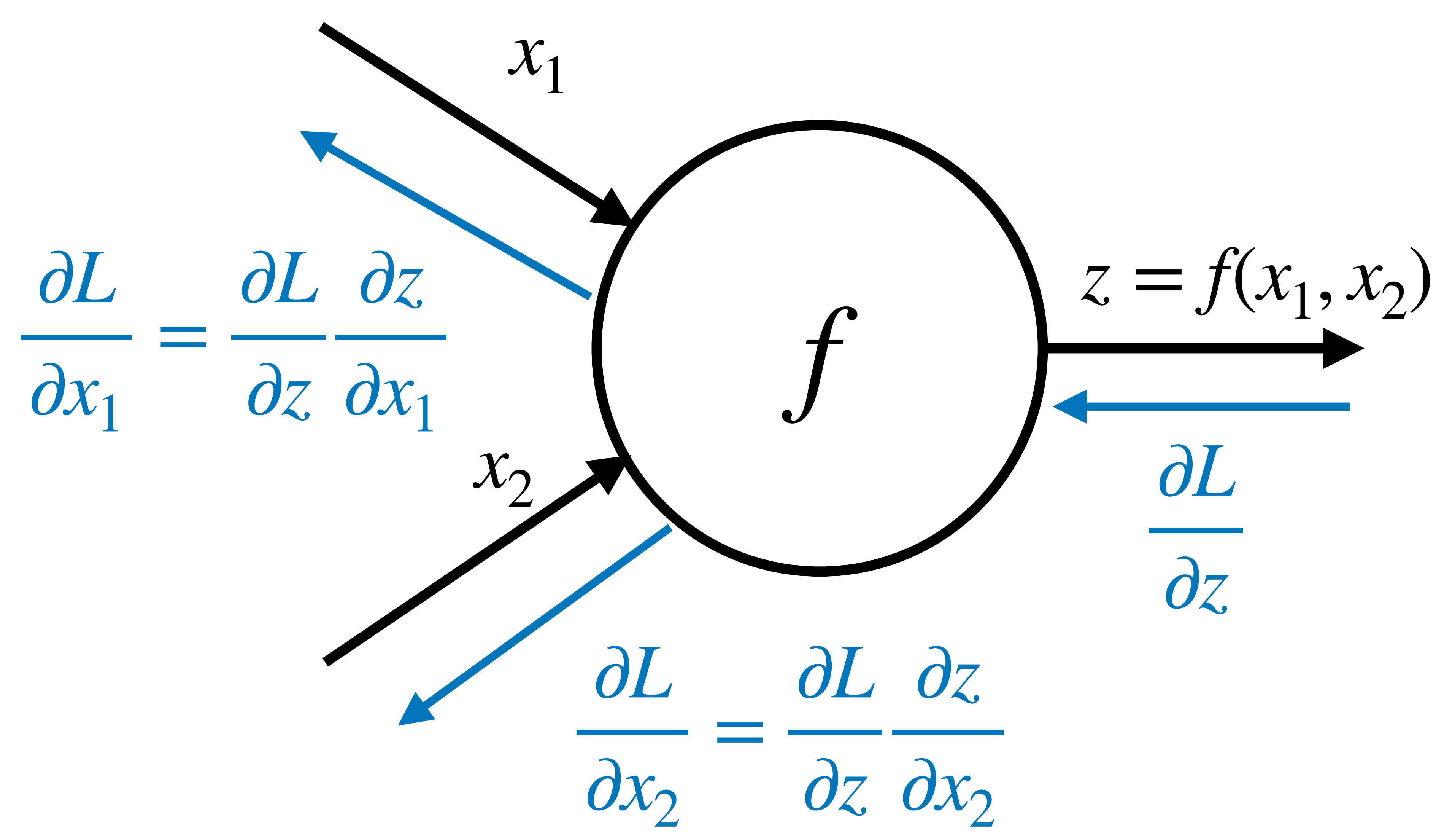
$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial h}\frac{\partial h}{\partial z} \qquad \frac{\partial h}{\partial z} = \frac{\partial f(z)}{\partial z} \qquad \frac{\partial L}{\partial h}$$

Downstream gradient = Local gradient × Upstream gradient

Gradient wrt input

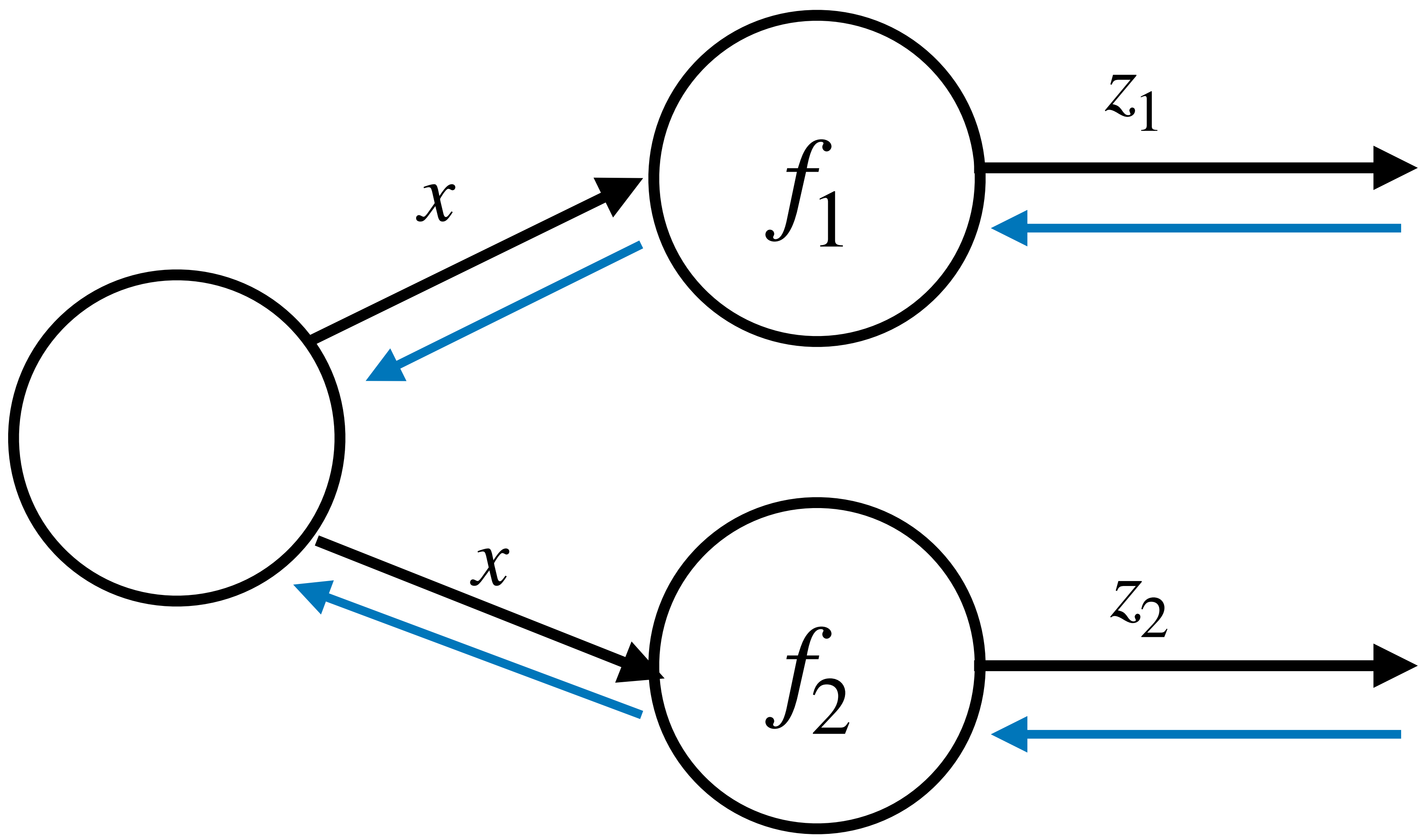
Backpropagation

Multiple inputs



Compute gradients for each input

Multiple output branches



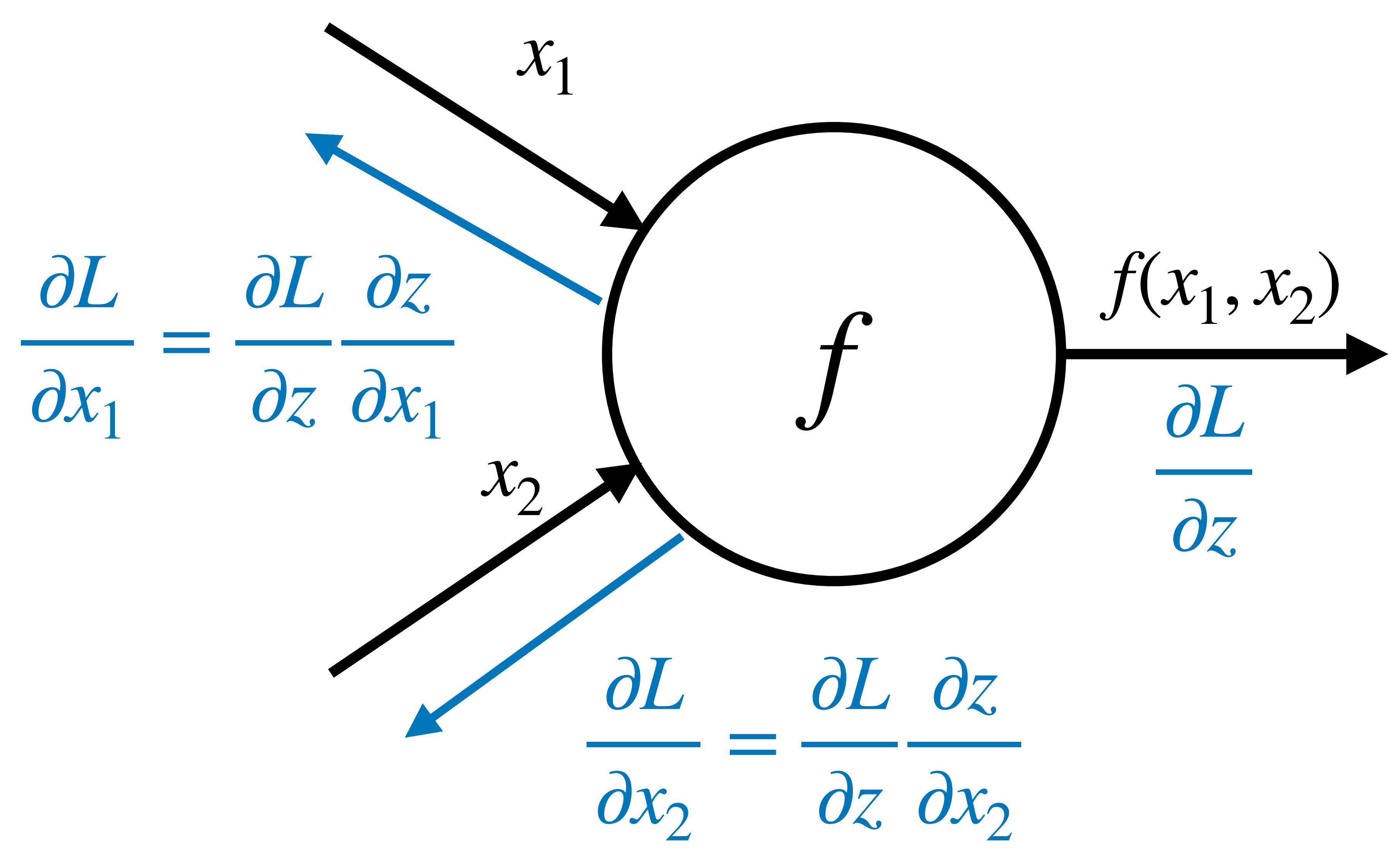
Sum gradients of branches

$$\frac{\partial L}{\partial x} = \sum_{i=1}^{n} \frac{\partial L}{\partial z_i}\frac{\partial z_i}{\partial x}$$

$$\{z_1, \ldots, z_n\} = \text{ successors of } x$$

Backpropagation API

Multiple inputs



Each node (operator) implement local forward/backward API

- forward(inputs)

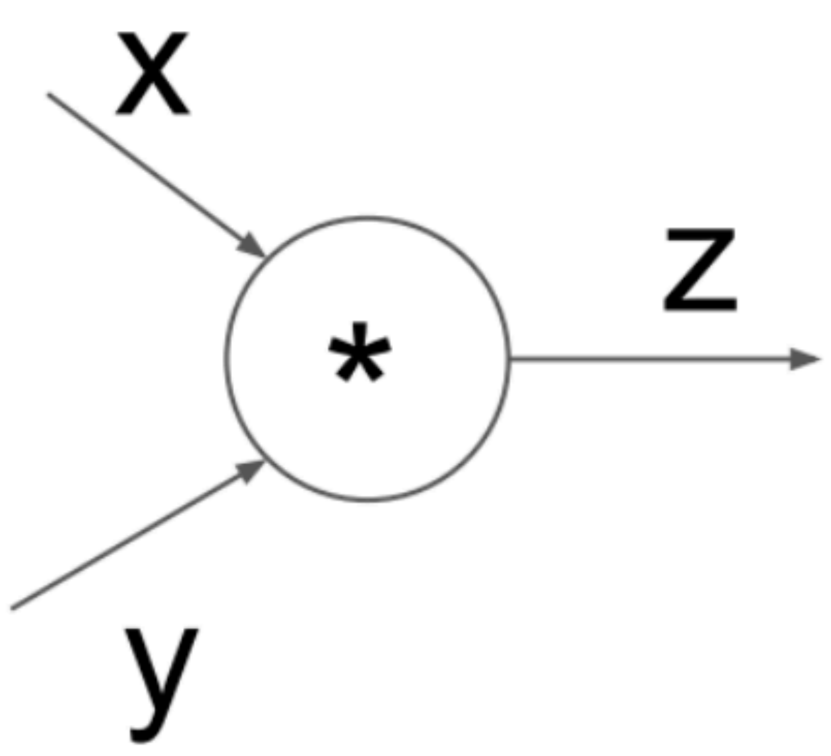
  $f(x_1, \ldots, x_k)$

- backward(upstream gradient)

  $\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_k}$

- Chain them together to form computation graph
- Reuse derivatives to minimize computation

Example: MultiplyGate



(x,y,z are scalars)

```python
class MultiplyGate(object):
    def forward(x,y):
        z = x*y
        self.x = x # must keep these around!
        self.y = y
        return z
    def backward(dz):
        dx = self.y * dz # [dz/dx * dL/dz]
        dy = self.x * dz # [dz/dy * dL/dz]
        return [dx, dy]
```

http://web.stanford.edu/class/cs224n/readings/cs224n-2019-notes03-neuralnets.pdf

Credits: Chris Manning (Stanford cs224n)

Backpropagation in general computational graph

- **Forward propagation**: visit nodes in topological sort order
  - Compute value of node given predecessors
- **Backward propagation**:
  - Initialize output gradient as 1
  - Visit nodes in reverse order and compute gradient wrt each node using gradient wrt successors

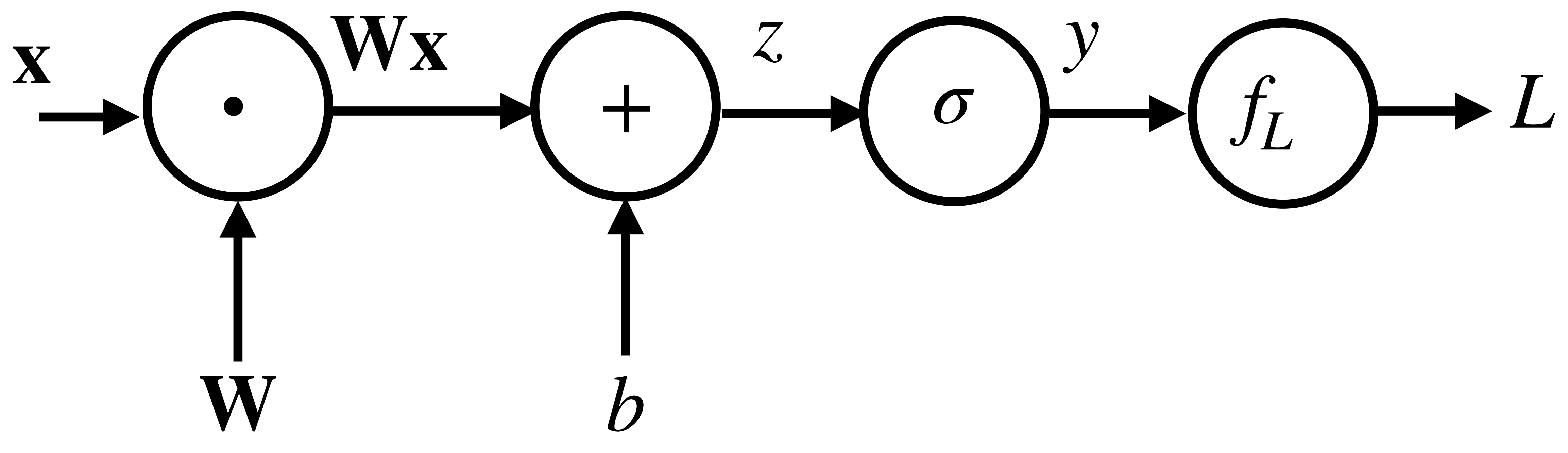
$$\frac{\partial L}{\partial x} = \sum_{i=1}^{n} \frac{\partial L}{\partial z_i} \frac{\partial z_i}{\partial x}$$

$$\{z_1, \ldots, z_n\} = \text{ successors of } x$$

For more details see https://colah.github.io/posts/2015-08-Backprop/

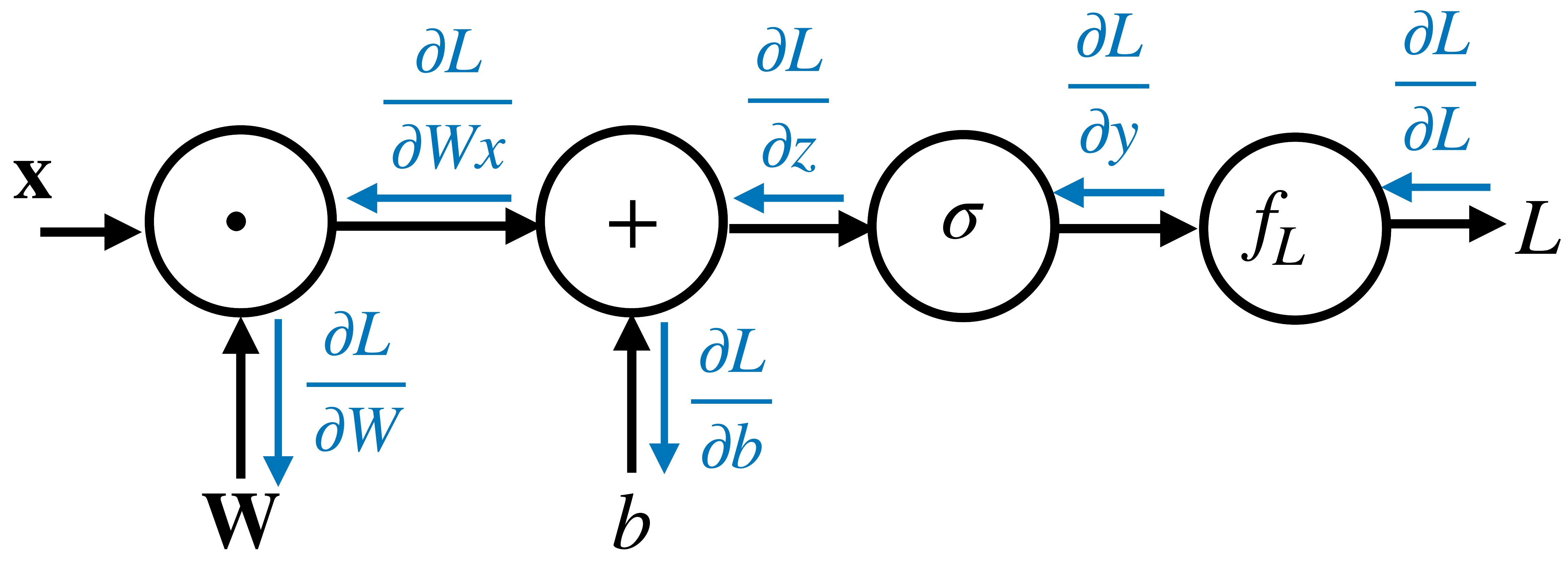
Simplified example

Forward pass: compute function value



$$\mathcal{L}(y, y^*) = -y^* \log y - (1 - y^*) \log (1 - y)$$

Backward pass: gradient using chain rule



Chain rule

$$\frac{\partial L}{\partial L} = 1$$

$$\frac{\partial L}{\partial y} = \frac{\partial L}{\partial L}\frac{\partial f_L(y, y^*)}{\partial y}$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial z} = \frac{\partial L}{\partial y}\frac{\partial \sigma(z)}{\partial z}$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial z}\frac{\partial z}{\partial b} = \frac{\partial L}{\partial z}\frac{\partial (\mathbf{Wx} + b)}{\partial b}$$

$$\frac{\partial L}{\partial \mathbf{Wx}} = \frac{\partial L}{\partial z}\frac{\partial z}{\partial \mathbf{Wx}} = \frac{\partial L}{\partial z}\frac{\partial (\mathbf{Wx} + b)}{\partial \mathbf{Wx}}$$

$$\frac{\partial L}{\partial \mathbf{W}} = \frac{\partial L}{\partial \mathbf{Wx}}\frac{\partial \mathbf{Wx}}{\partial \mathbf{W}}$$

Local derivatives

$$\frac{\partial f_L(y, y^*)}{\partial y} = y - y^*$$

$$\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1 - \sigma(z))$$

$$\frac{\partial (\mathbf{Wx} + b)}{\partial b} = 1$$

$$\frac{\partial (\mathbf{Wx} + b)}{\partial \mathbf{Wx}} = 1$$

$$\frac{\partial \mathbf{Wx}}{\partial \mathbf{W}} = \mathbf{x}^T$$

An example

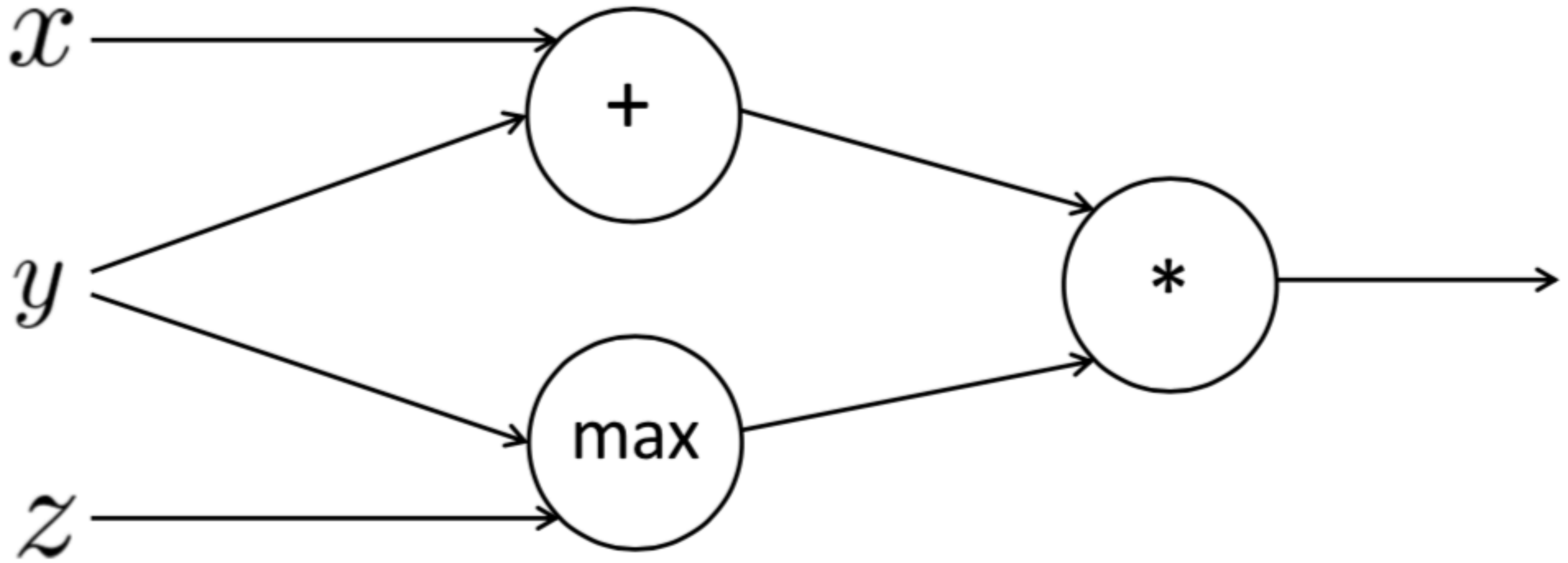
Try to compute the gradients yourself!

$$\boxed{\begin{array}{l} f(x, y, z) = (x + y)\max(y, z) \\ x = 1, y = 2, z = 0 \end{array}}$$

$$a = x + y$$

$$b = \max(y, z)$$

$$f = ab$$

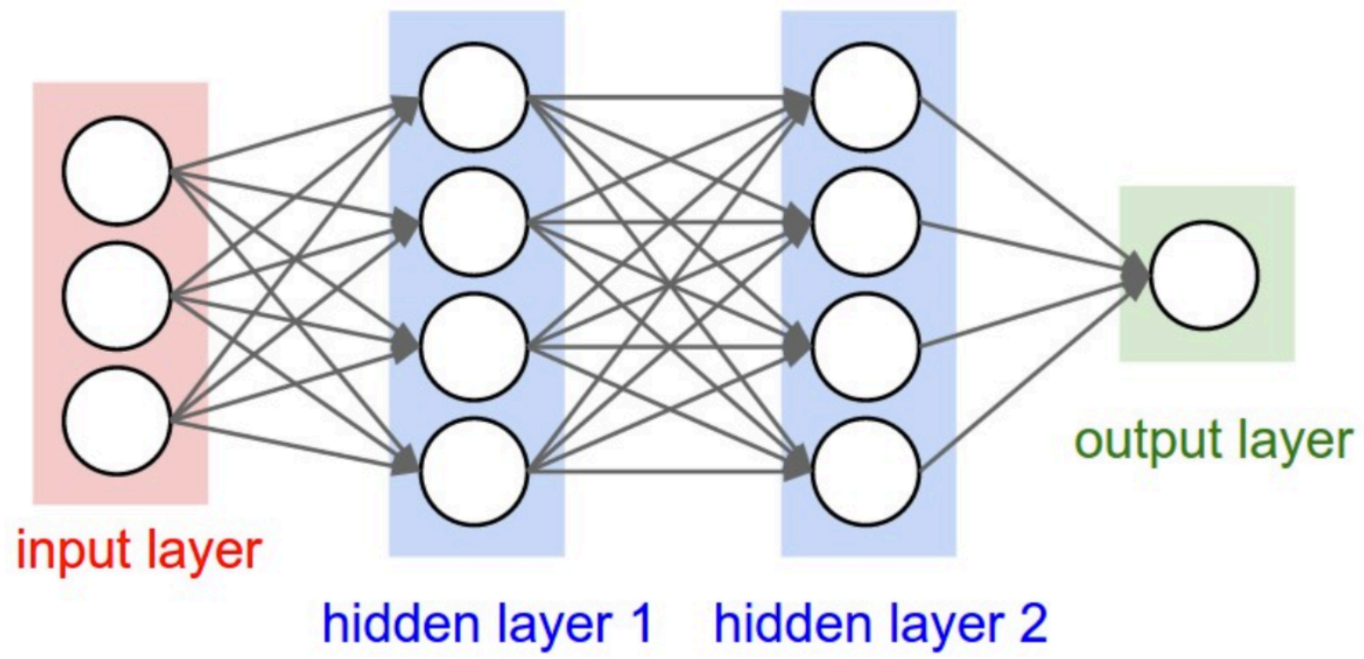


http://web.stanford.edu/class/cs224n/readings/cs224n-2019-notes03-neuralnets.pdf

Designing classifiers
with neural networks

Feature design: partly eliminated,
partly rolled into network design

- Input **features**: $f(x) \rightarrow [f_1, f_2, \ldots, f_m]$
  - Need to determine features
- Output: estimate $P(y = c \mid x)$ for each class c
  - Need to model $P(y = c \mid x)$ with a **family of functions**
- Train phase: Learn **parameters** of model to minimize loss function
  - Need **Loss function** and **Optimization** algorithm
- Test phase: Apply parameters to predict class given a new input



Neural Networks
figure out architecture

Still need to figure out loss function

General methods using auto-differentiation

Rise of deep-learning frameworks

Pytorch, TensorFlow, Keras, Theano, …

Provide frequently used components that can be connected together

- Easy to build complex models
  - Connect up neural building blocks
  - Mix and match selection of loss functions, regularizers, and optimizers
- Optimize using auto-differentiation, no need to hand code optimizers for specific models
- Deals with numerical stability issues
- Deals with efficient computation (e.g. batching, using GPUs)
- Provides (some) experiment logging and visualization tools

No longer need to code all the pieces of your model and optimizer by yourself

Allows researchers and developers to focus on

- Modeling the problem
- Designing the network

Resources

- There is a lot more to learn about neural networks!
  - TA Tutorials
    - Optimization
    - Pytorch and backpropagation
    - Debugging Tips and Tricks
  - Classes: CMPT 728 (Deep Learning)

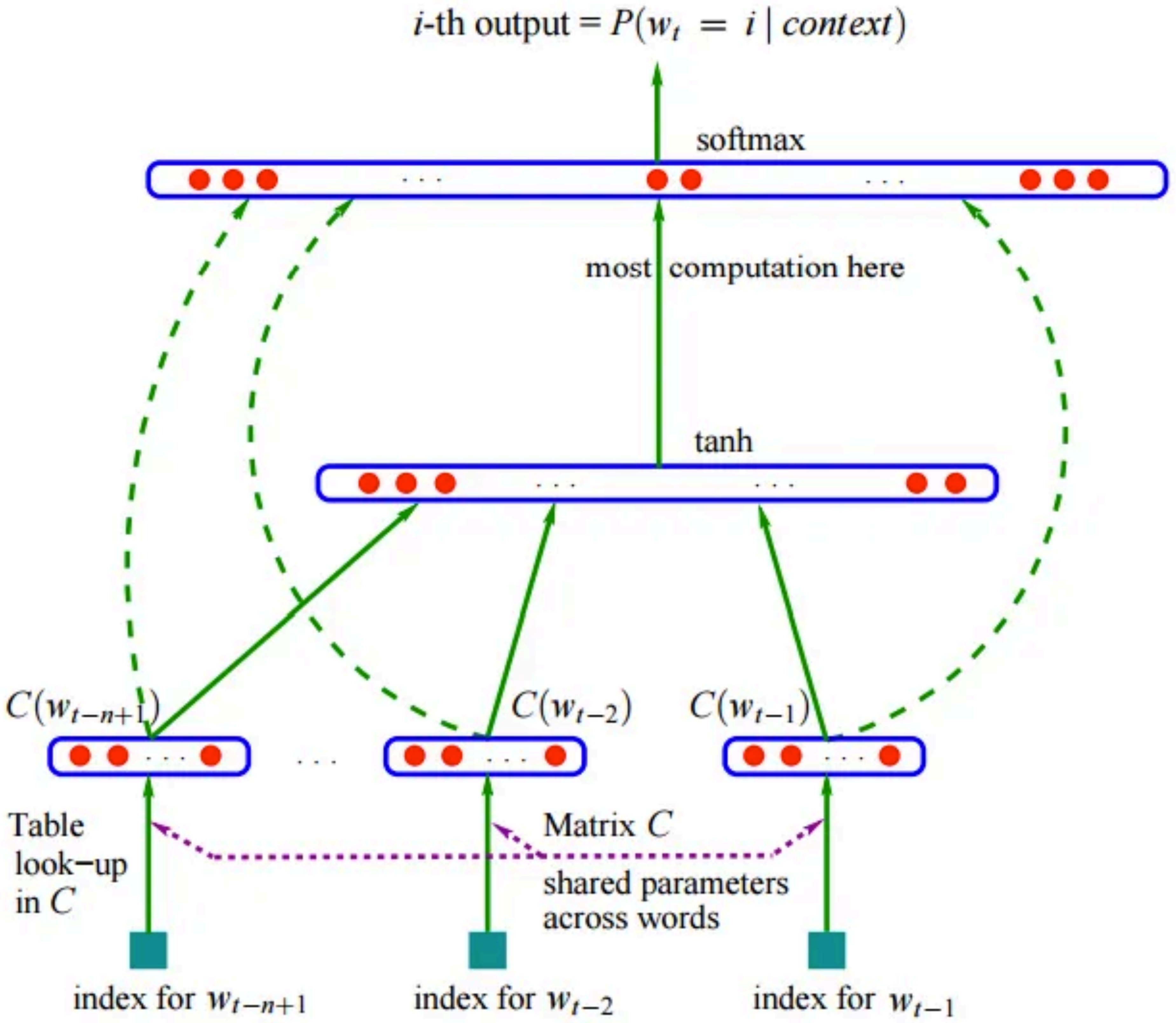
Resources

- Deep learning books
  - Courville, Goodfellow, and Bengio: https://www.deeplearningbook.org/
  - Yoav Goldberg's Primer on NN models for NLP: http://u.cs.biu.ac.il/~yogo/nnlp.pdf
- Classes from other universities
  - Stanford CS231n notes: https://cs231n.github.io/
  - University of Toronto: https://csc413-2020.github.io/ (Jimmy Ba and Roger Grosse)
  - University of Michigan: https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/ (Justin Johnson)

Applications

Feedforward Neural LMs

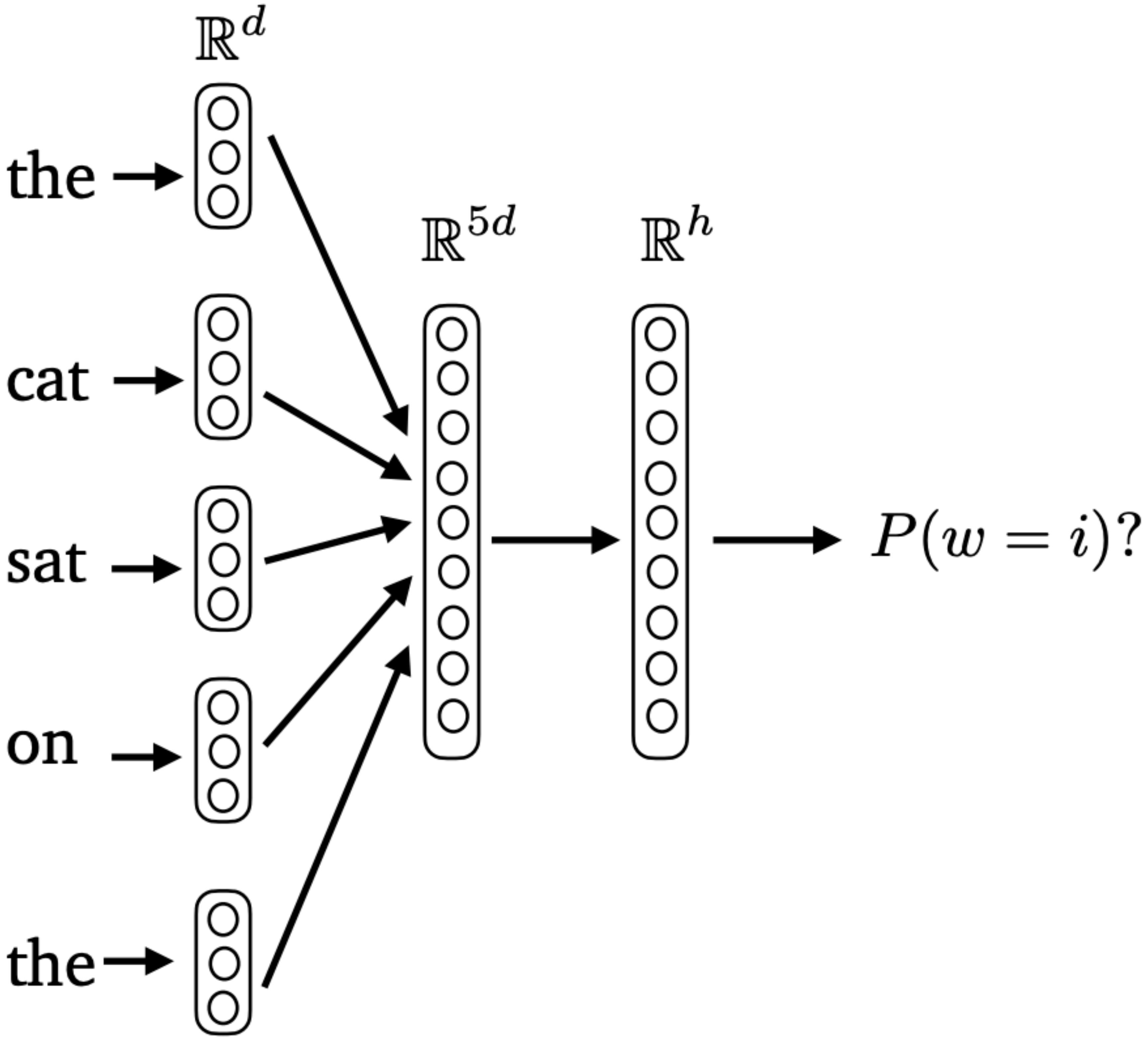
- N-gram models: $P(\text{mat}|\text{the cat sat on the})$



(Bengio et 2003): A Neural Probabilistic Language Model

Feedforward Neural LMs

- P(mat | the cat sat on the) = ?



- Input layer (context size n = 5):

$$\mathbf{x} = [\mathbf{e}_{\text{the}}; \mathbf{e}_{\text{cat}}; \mathbf{e}_{\text{sat}}; \mathbf{e}_{\text{on}}; \mathbf{e}_{\text{the}}] \in \mathbb{R}^{dn}$$

concatenation

- Hidden layer

$$\mathbf{h} = \tanh(\mathbf{W}\mathbf{x} + \mathbf{b}) \in \mathbb{R}^h$$

- Output layer (softmax)

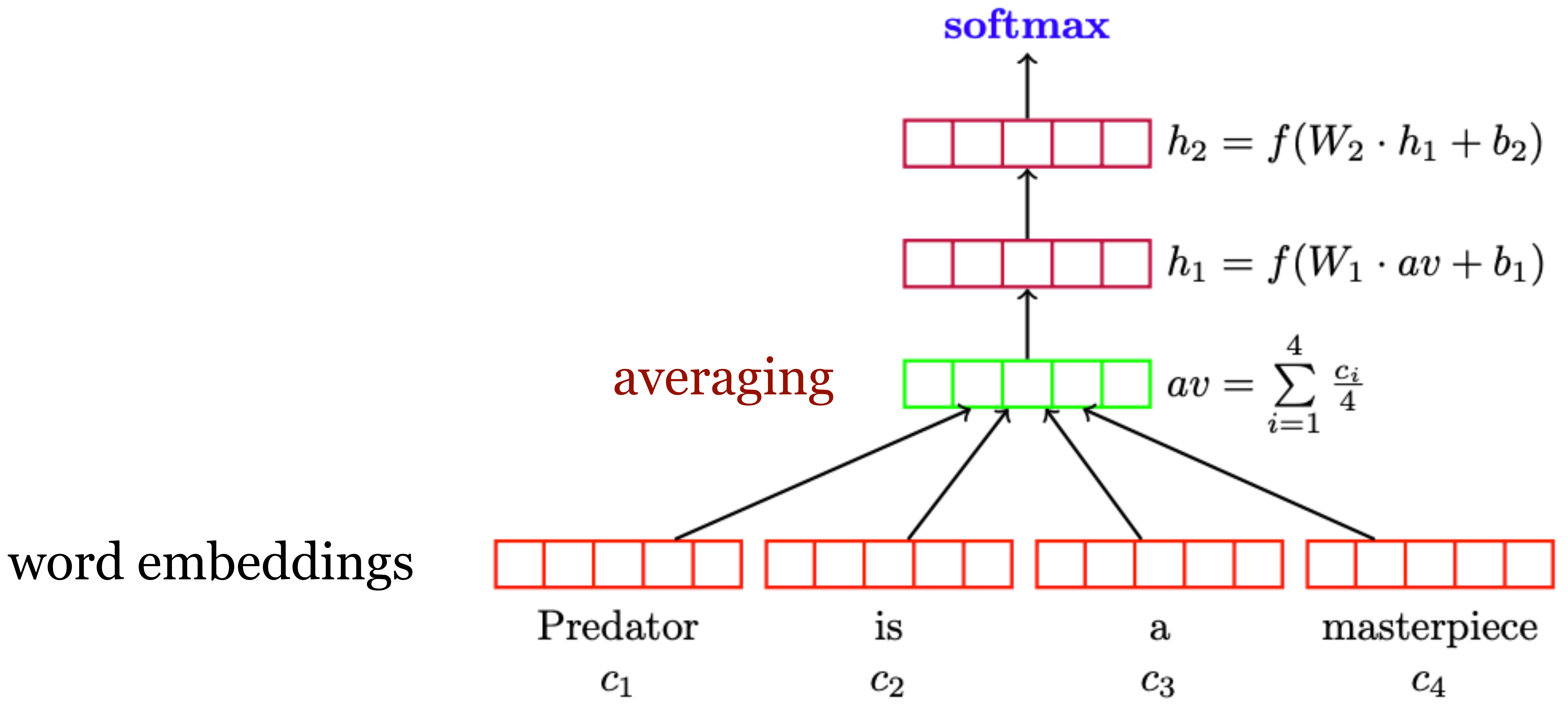
$$\mathbf{z} = \mathbf{U}\mathbf{h} \in \mathbb{R}^{|V|}$$

$$P(w = i \mid \text{context}) = softmax_i(\mathbf{z})$$

(Bengio et 2003): A Neural Probabilistic Language Model

Neural Bag-of-Words (NBOW)

- Deep Averaging Networks (DAN) for Text Classification



(Iyyer et 2015): Deep Unordered Composition Rivals Syntactic Methods for Text Classification